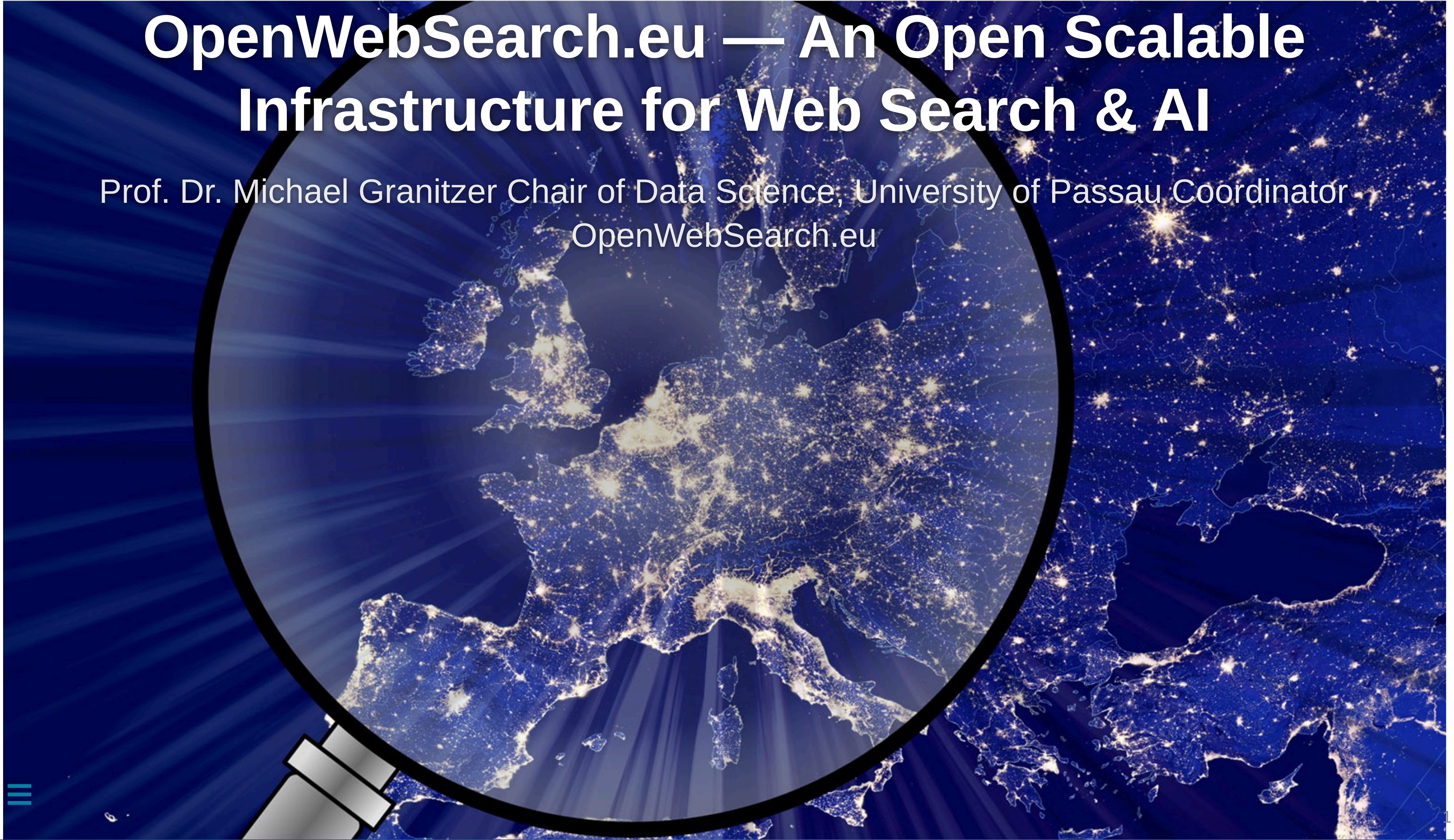
OpenWebSearch.eu — An Open Scalable Infrastructure for Web Search & AI

Prof. Dr. Michael Granitzer Chair of Data Science, University of Passau Coordinator OpenWebSearch.eu

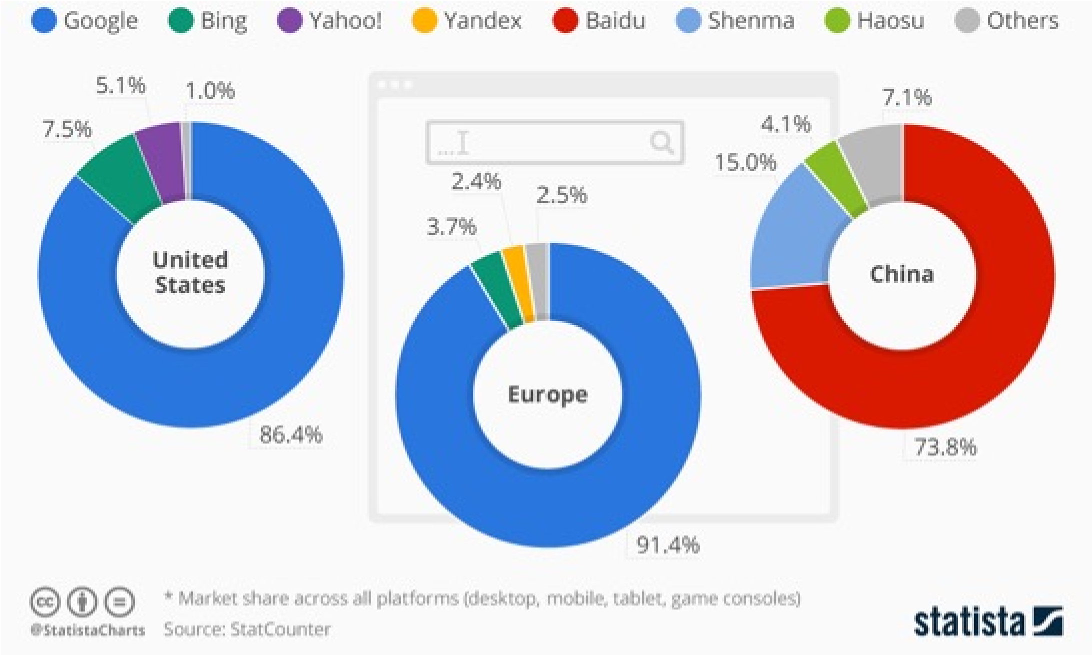


Web Search Today

A critical resource managed as an oligopoly

Two properties that don't fit

- A **critical infrastructure** for society, comparable to satellite navigation
- A **market oligopoly**: dominated by a small number of gatekeepers (Google, Microsoft, Baidu, ...)



But does it matter? Yes — on four dimensions

User Choice & Democracy

- Single gatekeeper for society's information
- Search rankings can shift voting preferences by **20–72%** (SEME)
- Monopolistic ranking = single point of failure for information quality

Societal & Security Risks

- European OSINT depends on US-controlled infrastructure
- Filter bubbles, algorithmic bias
- Misinformation & Disinformation

Untapped Richest Information Source

- 60 % of web pages carry structured data
- No open access to web-scale document collections
- Science, research and technology heavily depend on the Web as platform

AI & Innovation

- Web data is the fuel for LLMs, RAG, and agentic AI
- Closed indices block reproducible research & open innovation

> Being able to navigate the Web is as fundamental as GPS or the power grid

Challenges

THE WEB: VOLUME, VELOCITY, VARIETY

Careful: domains, sites, pages, and indexed documents are different units

VOLUME

1.15 Billion

Netcraft site responses

272.6 M domains · Dec 2024 Netcraft survey

Top 0.1%–1% of sites

may account for roughly half of all pages

heavy-tail intuition from large web graph studies

~ 400 Billion

documents in Google’s search index

2020 figure cited in US v. Google testimony

Sites ≠ domains ≠ pages; use exact counts cautiously

VELOCITY

+8.6 M / month

monthly net change in Netcraft site responses

Dec 2024 versus Nov 2024 Netcraft survey

~ 3.2 / second

same monthly net change, averaged across December

149 ZB (global)

all digital data created / copied worldwide in 2024

IDC / Statista global datasphere estimate

The web changes constantly; crawling is continuous, not one-off

VARIETY

150+ languages

observed across websites

W3Techs usage survey

249 CMS Platforms

CMS platforms identified in the HTTP Archive dataset

HTTP Archive Web Almanac 2024 (Wappalyzer-based)

HTML · PDF · JSON-LD · media · APIs

content types to parse and process

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Sources: Netcraft Web Server Survey Dec 2024 · large web graph studies (heavy-tail intuition) ·
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- High infrastructure costs
- Broad range of technical skills required
  - Big Data Infrastructure
  - Natural Language Processing, Image Analysis
  - Web Technologies
- Growing legal uncertainties

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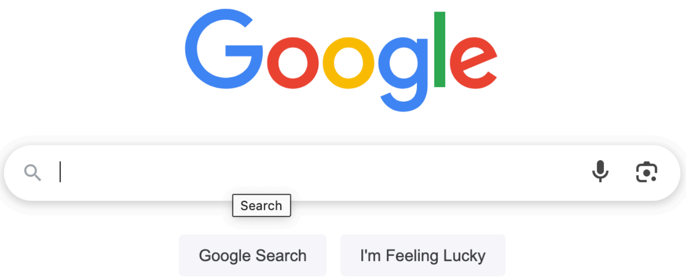
- High infrastructure costs
- Broad range of technical skills required
  - Big Data Infrastructure
  - Natural Language Processing, Image Analysis
  - Web Technologies
- Growing legal uncertainties

High upfront costs to tap the Web as resources — especially small innovators and researchers are left behind

WHY IT MATTERS - IN MORE DETAILS

Reduced user choice

Single source of our information: We would not be happy with only one European newspaper



Newspaper map: reddit.com/r/MapPorn/comments/1c70v2q/newspapers_of_record_or_note_in_europe/

Search Engine Manipulation Effect

- Users locked into a single information gatekeeper
- **Search Engine Manipulation Effect (SEME)**: biased search rankings can shift voting preferences of undecided voters by **20% or more** — up to **72%** in some demographics — while users remain largely unaware of the manipulation

Epstein, R. & Robertson, R.E. "The search engine manipulation effect (SEME) and its possible impact on the outcomes of elections." PNAS 112(33), E4512–E4521, 2015. DOI: 10.1073/pnas.1419828112

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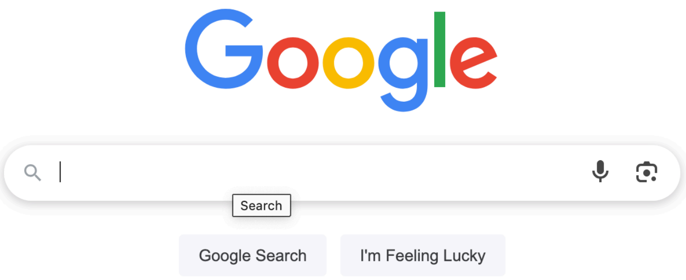

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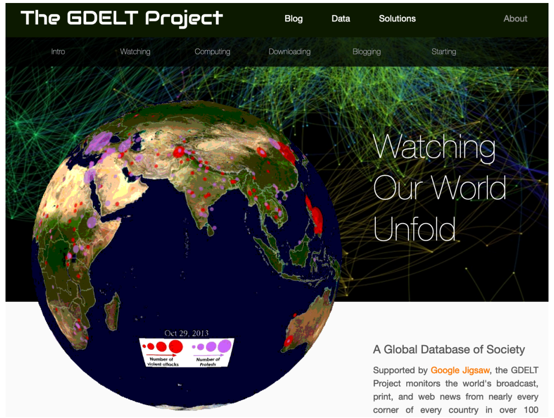
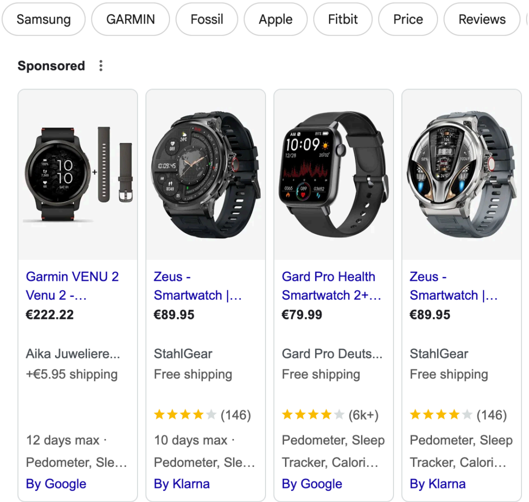
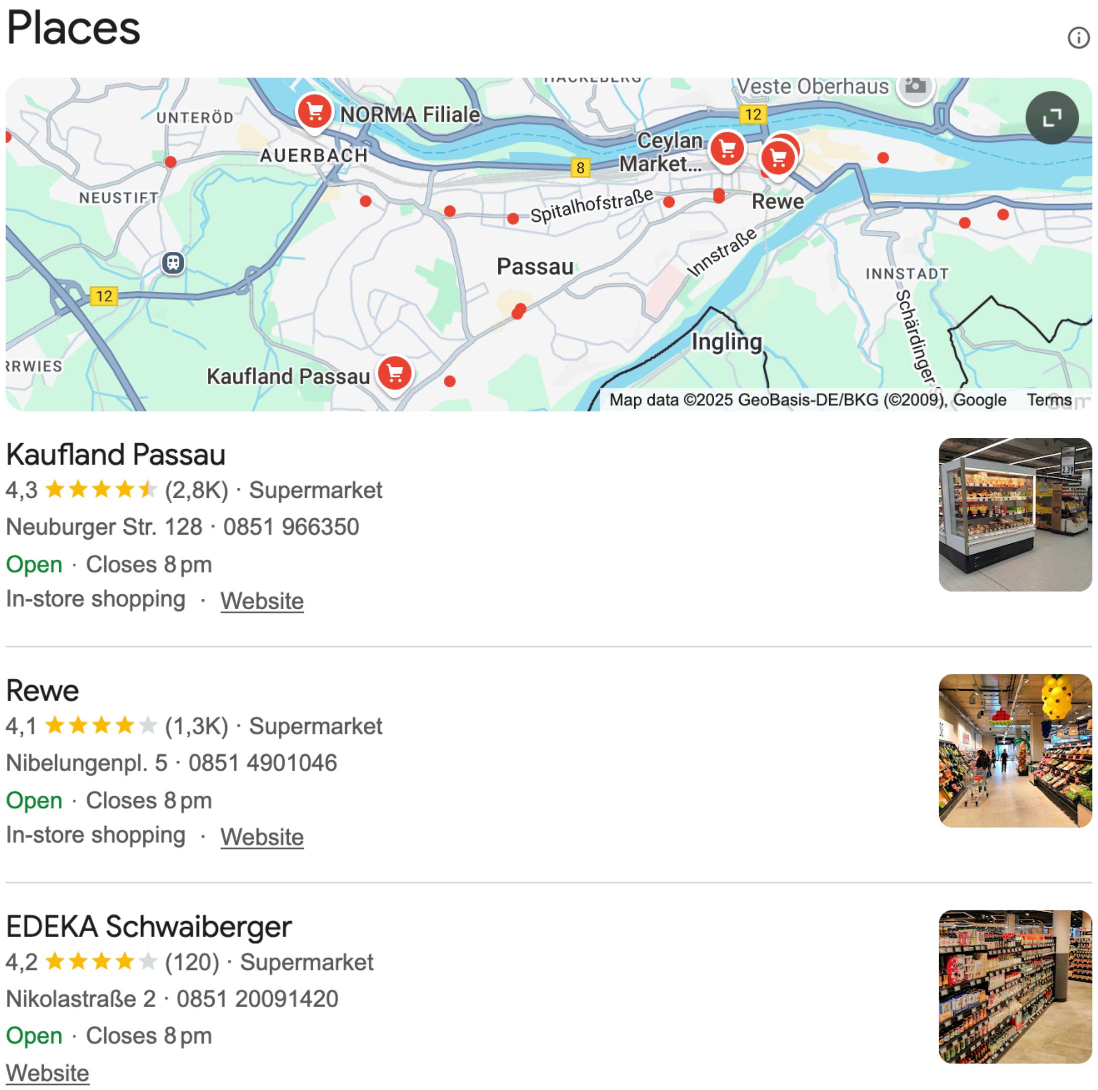
The web is the richest information source

The Web is the richest information source, but we are blind on what is there

- Companies, products, scientific information, events, geo-location data
- 60% of web pages contain structured data in form of Microdata / Microformats
    - Products, FAQ, Places, Shops
    - Job postings, Reviews, Events







Societal Challenges which require a grip on the Web

Misinformation / Disinformation

- Monopolistic ranking creates single points of failure for information quality
- Lack of index transparency prevents independent auditing and cross-language disinfo detection
- Open indices enable reproducible fact-checking pipelines

Guidance & Algorithmic Bias

- Search autocompletions and knowledge panels shape health, legal and consumer decisions
- Filter bubbles create asymmetric information through personalization
- Low-income users disproportionately affected by low-quality results

Defence & Security

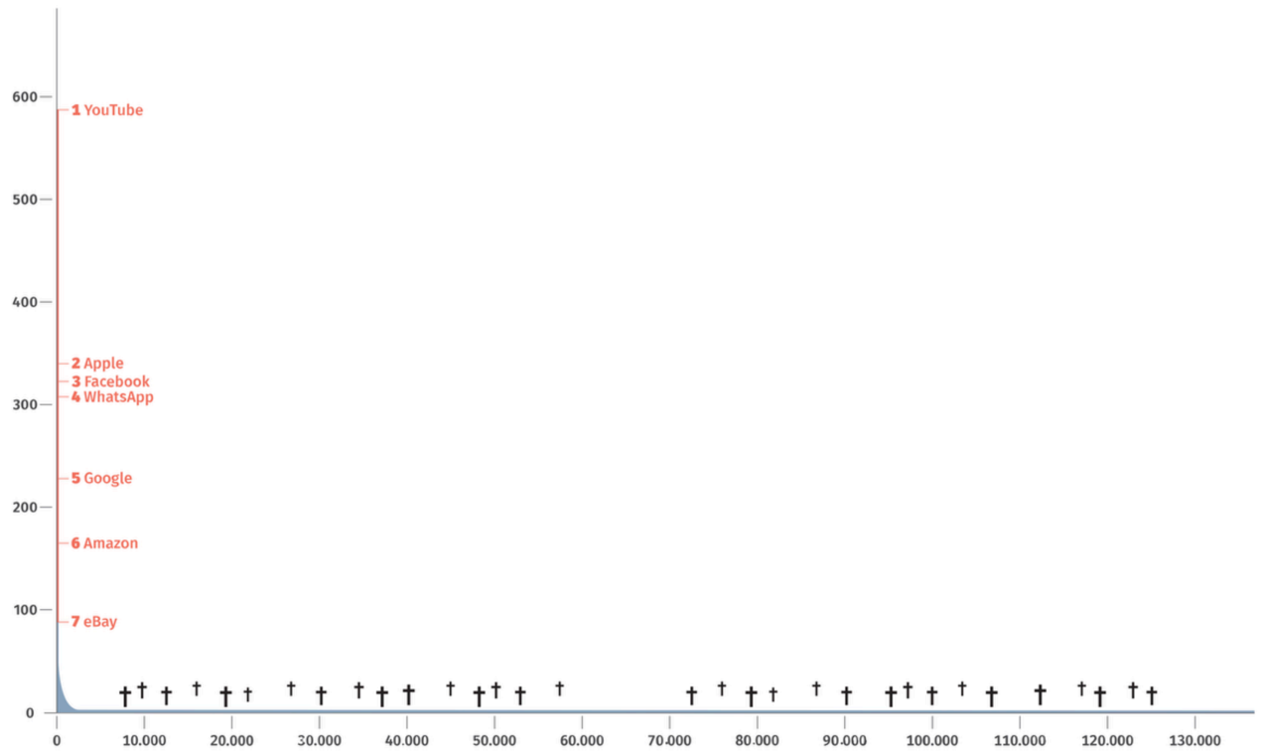
- European OSINT depends on US-controlled indices — strategic vulnerability in crisis scenarios
- Adversarial SEO ("search engine poisoning") is a documented hybrid warfare tactic
- No sovereign search infrastructure for independent web monitoring

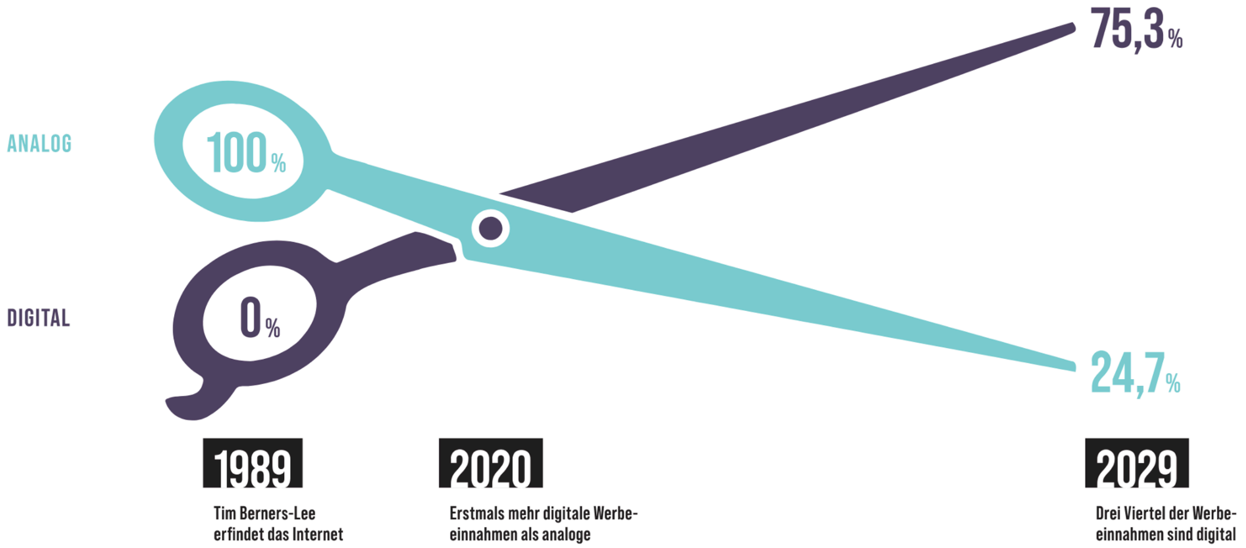
Digital & Information Literacy

- Most users cannot distinguish ads from organic results — "trust by ranking" effect
- Algorithmic literacy remains low even among university students
- Transparent, inspectable search infrastructure is a prerequisite for teaching information literacy effectively

Europe is not present in the web

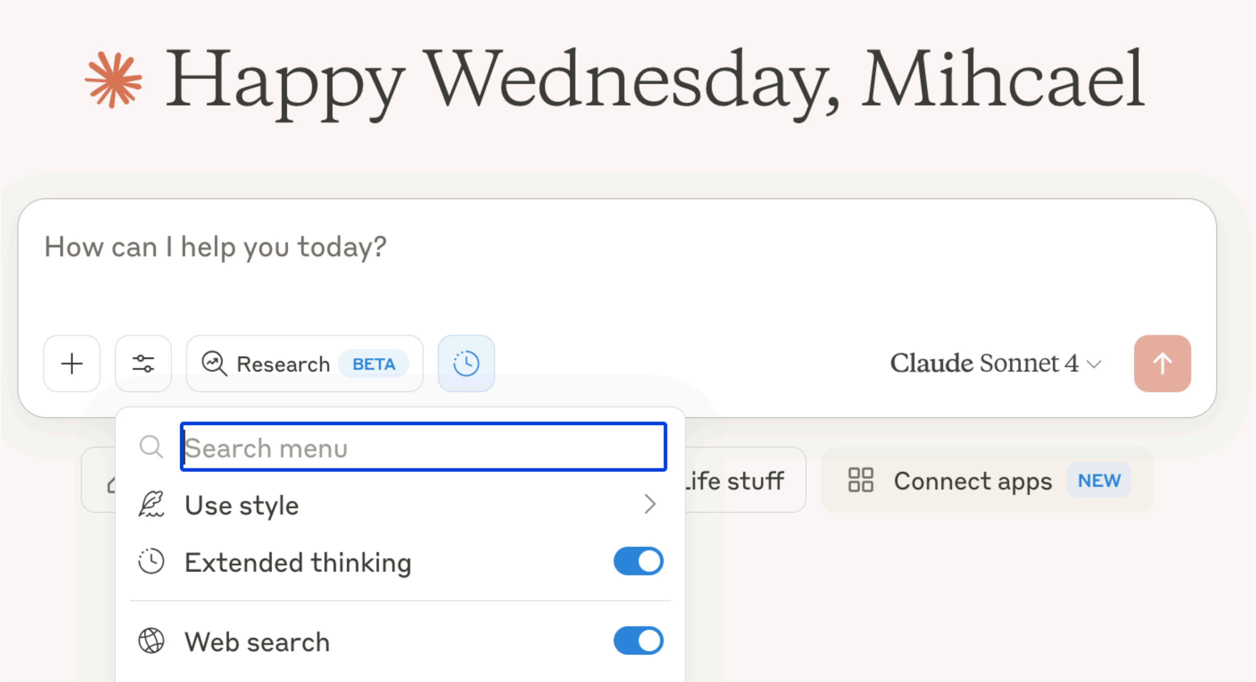
- Advertising revenue consolidated by US gatekeepers
- European content, languages, and perspectives underrepresented in search results
- No European alternative for web-scale data access

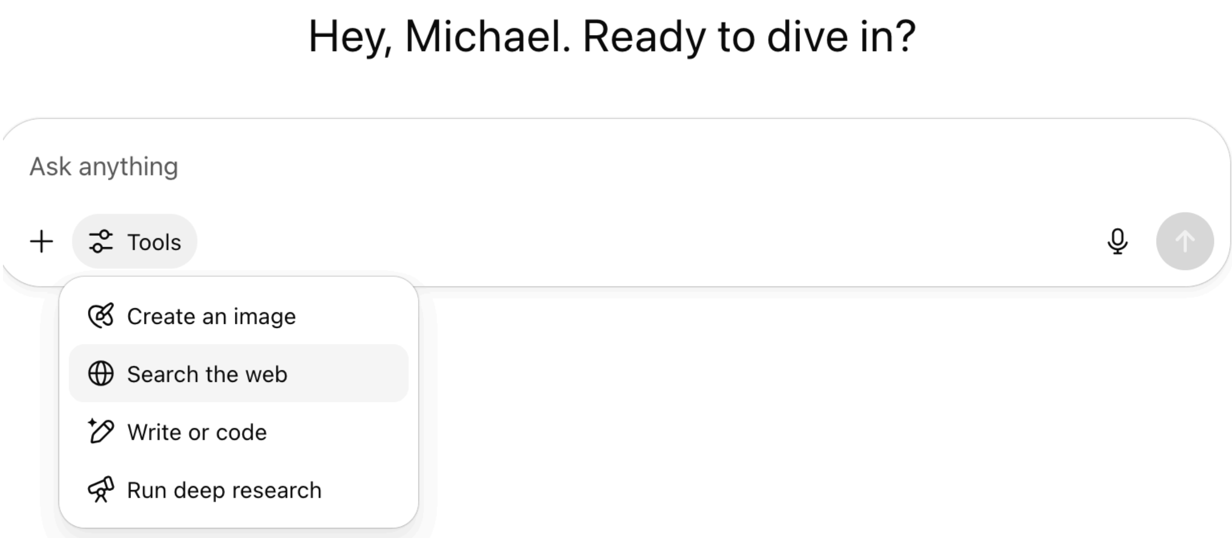




The web drives innovation and AI

- Web data is essential for training LLMs and VLMs
- Retrieval Augmented Generation (RAG) requires web-scale document collections
- Agentic AI systems rely on web search as a core tool
- Web search remains an integral part of modern AI pipelines





OpenWebSearch.eu — Our Mission

> Building an Open Index of the Web and an open infrastructure to reduce upfront costs for researchers, innovators and companies in tapping the Web as a resource for search, web-analytics and AI.

Four Objectives

1. **Open Technology Stack** Open-source pipelines for crawling, preprocessing, indexing
2. **Resource Provision** Building a network of infrastructure providers (EuroHPC / AI Factories)
3. **Added Value Services** Dashboard, tools, data access APIs
4. **Bootstrapping the Ecosystem** Third-party calls, community building, applications and use-cases

14 Core Partners

Infrastructure

Research

NGOs

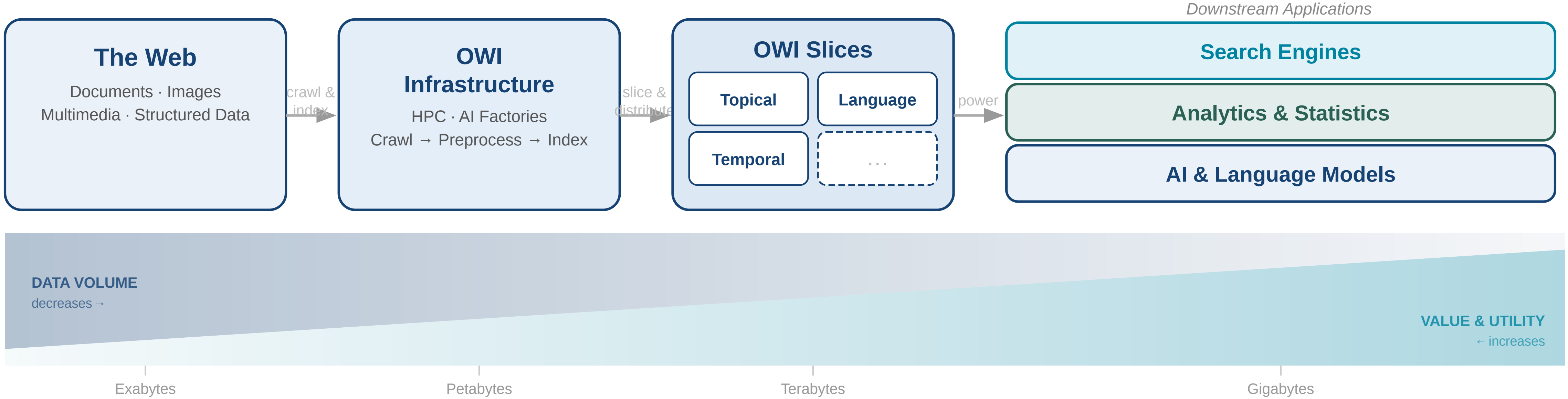
Businesses

Plus 16 financially supported third-party projects (FSTPss)

The Open Web Index (OWI)

A **Web Index** = a data structure for fast query-based access and ranking of web documents — the core of every web search engine

Our Proposition: A collaboratively created, open and transparent Web Index running on existing HPC computing centres / AI Factories in Europe



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What is the Open Web Index?

The Open Web Index

The Open Web Index — a federated, daily-published web index built across European HPC sites and consumed as composable shards.

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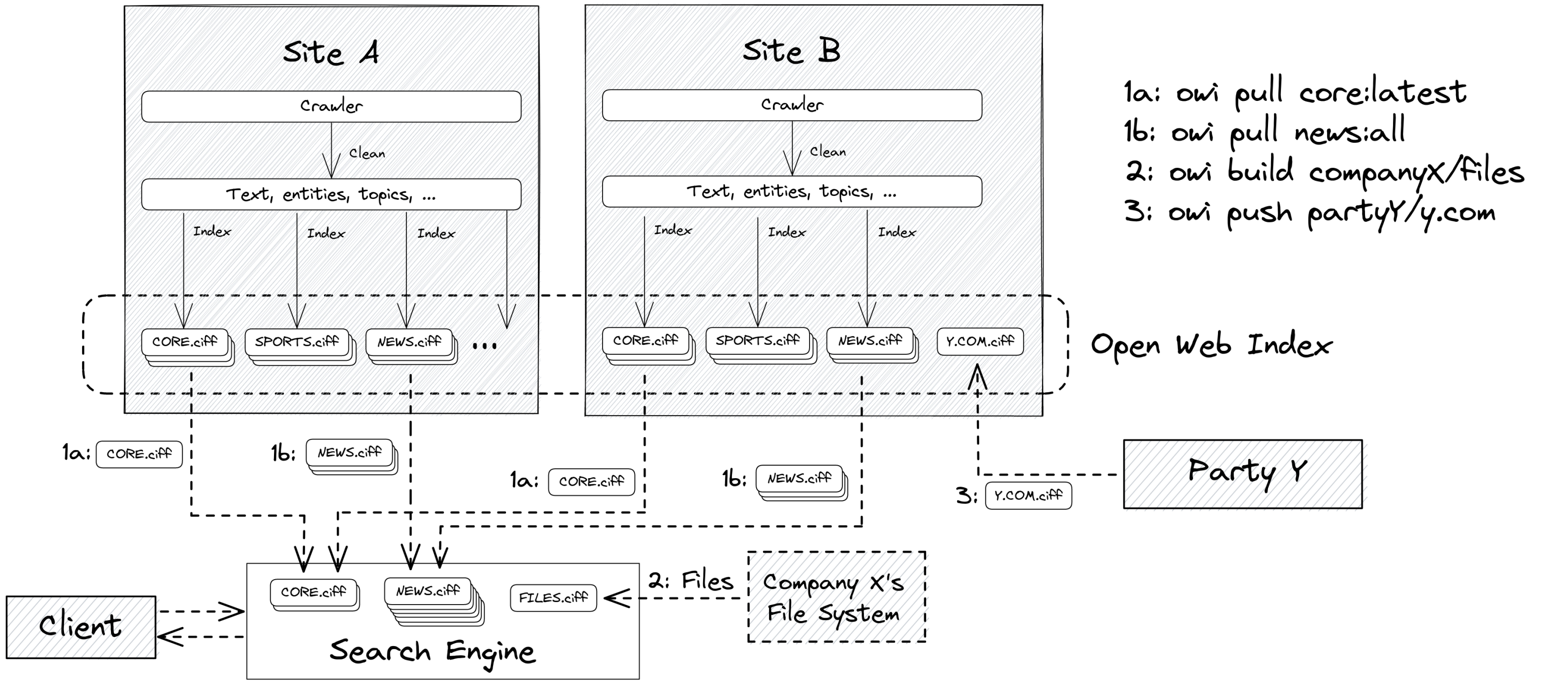


Federated production

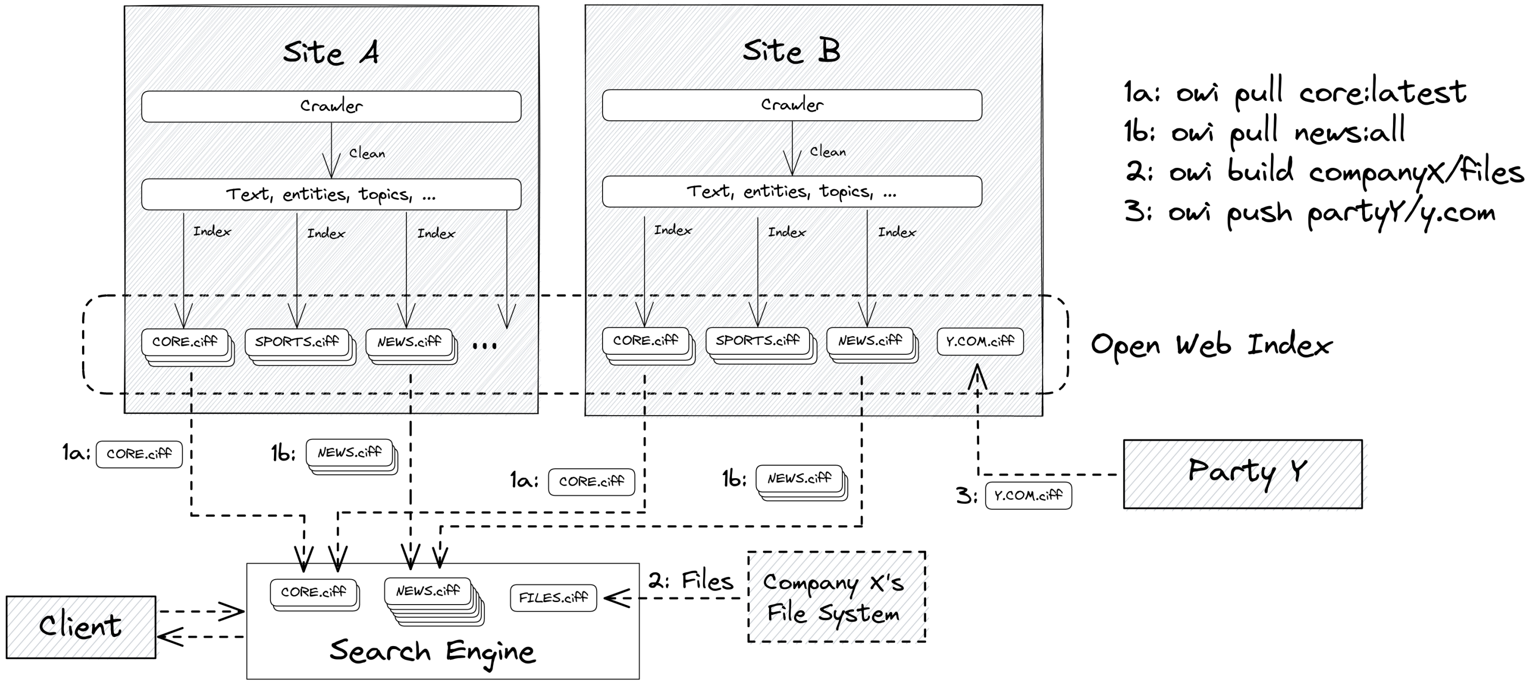
- each HPC site runs the same pipeline independently
- daily shards by date, language, topic

Three access verbs

- `owi pull` — download pre-built shards
- `owi build` — index your own data
- `owi push` — contribute shards back

The Open Web Index

The Open Web Index — a federated, daily-published web index built across European HPC sites and consumed as composable shards.



Federated production

- each HPC site runs the same pipeline independently
- daily shards by date, language, topic

Three access verbs

- `owi pull` — download pre-built shards
- `owi build` — index your own data
- `owi push` — contribute shards back

Three output types per daily run — the building blocks of a shard:

OWI at a Glance — Statistics (Feb 2026)

Crawling

Pages crawled	~39 B total, ~12 B unique
Unique hosts	~500 M (extrapolated)
Crawling effort	1,927 machine-days
Avg. throughput	~20 M URLs / day

Index

Documents indexed	~10.7 B
Unique domains	~75 M
Public datasets	511 main / 1,511 total
Index size	45 TB (main) / 50 TB total
Embeddings	119 M docs · 884 GB

Top languages: English (38.9%), German (7.2%), Spanish (6.4%), French (5.7%), Russian (5.2%)

Multimedia Objects: Roughly 2-3 quality images and 0.5 Videos per Web-Document

Comparison: To meet commercial index size, OWI would need to scale by a factor of 20× — factor 10–50× for multimedia

IN DETAIL

Index Format: CIFF + Parquet

CIFF (Common Index File Format)

A Protobuf schema for sparse inverted files — the OWI standard interchange format:

- **Header** — collection stats: document count, unique terms, avg. document length
- **Postings lists** — per term: document frequency + collection frequency + list of (docid, tf) pairs
- **Document records** — internal numeric ID ↔ external URL + document length

Designed so any existing search engine can import the data and start ranking immediately. Extensions proposed for OWI-specific metadata.

Parquet Metadata Files

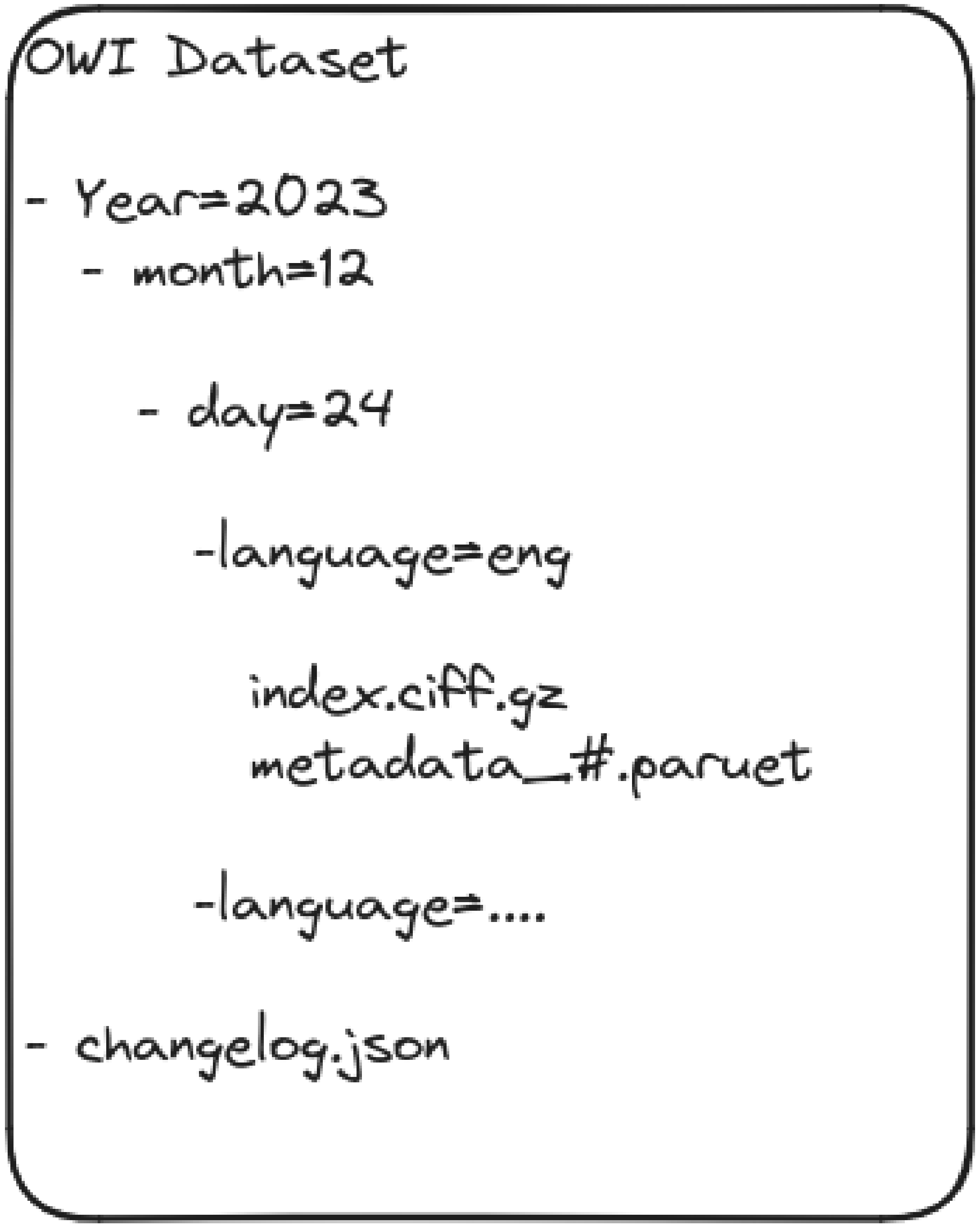
Distributed alongside each CIFF shard — aligned partitions containing:

- URL, title, language, crawl date
- Quality/importance scores from Resilipipe
- Topic / entity enrichments (WP 2)
- Enables SQL-style analytics without downloading the full index

Dense Embeddings (GPU slice)

- High-quality / high-importance doc subset
- Jina V3 embeddings, 884 GB in public datasets
- Enables semantic search and RAG on OWI content

Dataset Structure & Versioning



Each **daily shard** follows a consistent folder hierarchy:

```
OWI Dataset
  └── year=YYYY
        └── month=MM
              └── day=DD
                    ├── language=eng
                    │     ├── index.ciff.gz
                    │     └── metadata_*.parquet
                    ├── language=deu
                    │     └── ...
                    └── changelog.json
```

- One dataset per data center per day
- Collections: `main`, `curlie_full`, gpu (embeddings), and domain-specific slices
- `changelog.json` tracks document additions and deletions (take-down compliance)
- Git-like versioning via **owilix**: `pull`, `push`, `build`

Language Coverage

Top 15 Languages (main collection)

Rank	Language	Share
1	English	38.9%
2	German	7.2%
3	Spanish	6.4%
4	French	5.7%
5	Russian	5.2%
6	Japanese	4.9%
7	Chinese	3.5%
8	Italian	2.8%
9	Polish	1.6%

Collection Breakdown

Collection	Datasets	Size
main	511	45 TB
total (all)	1,511	50 TB
embeddings	58	884 GB

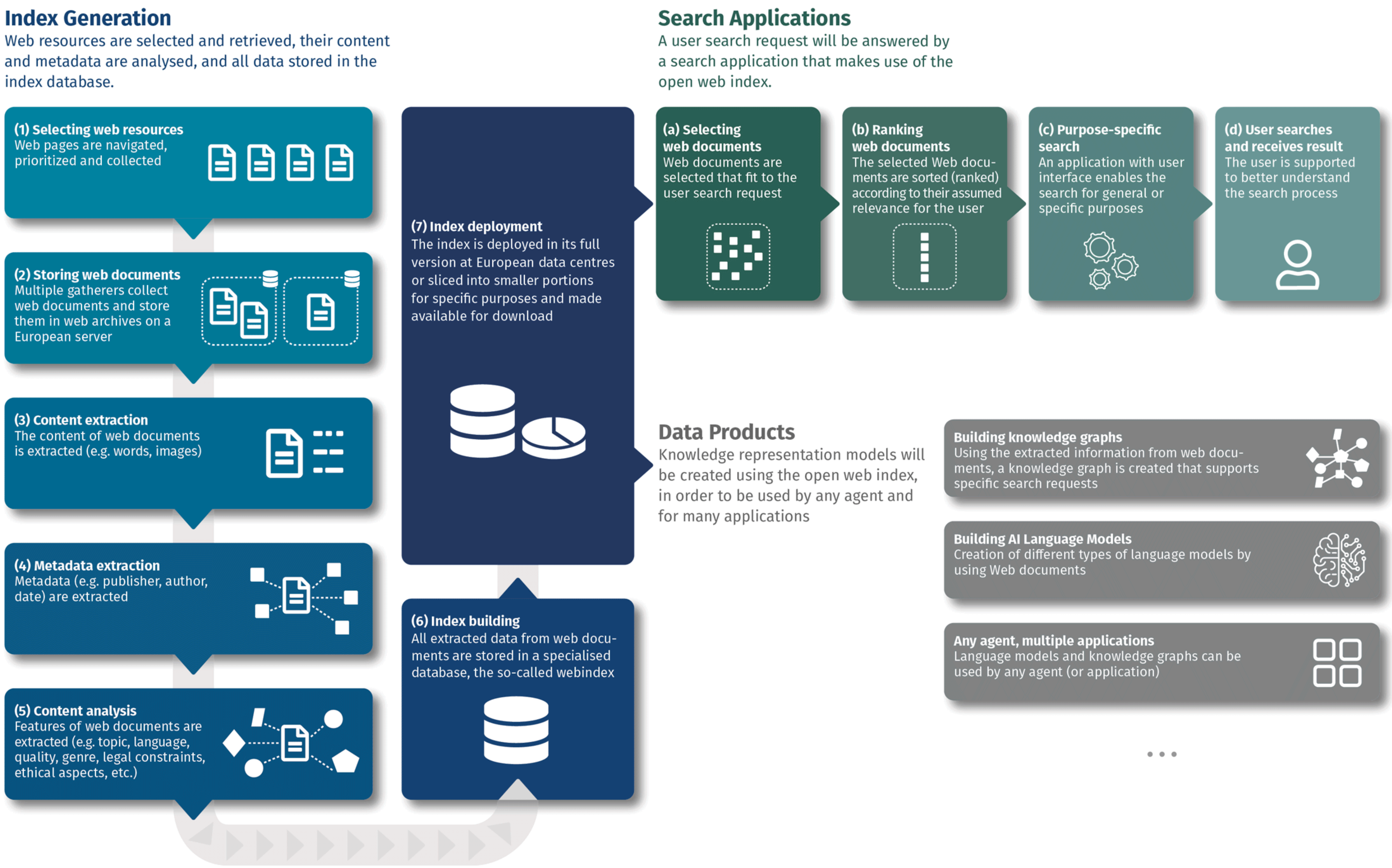
Dedicated language slices — individual CIFF shards per language enable language-specific downstream applications without full-index downloads

> All data, software and formats published as **open source** under permissive licences · Archived on Zenodo

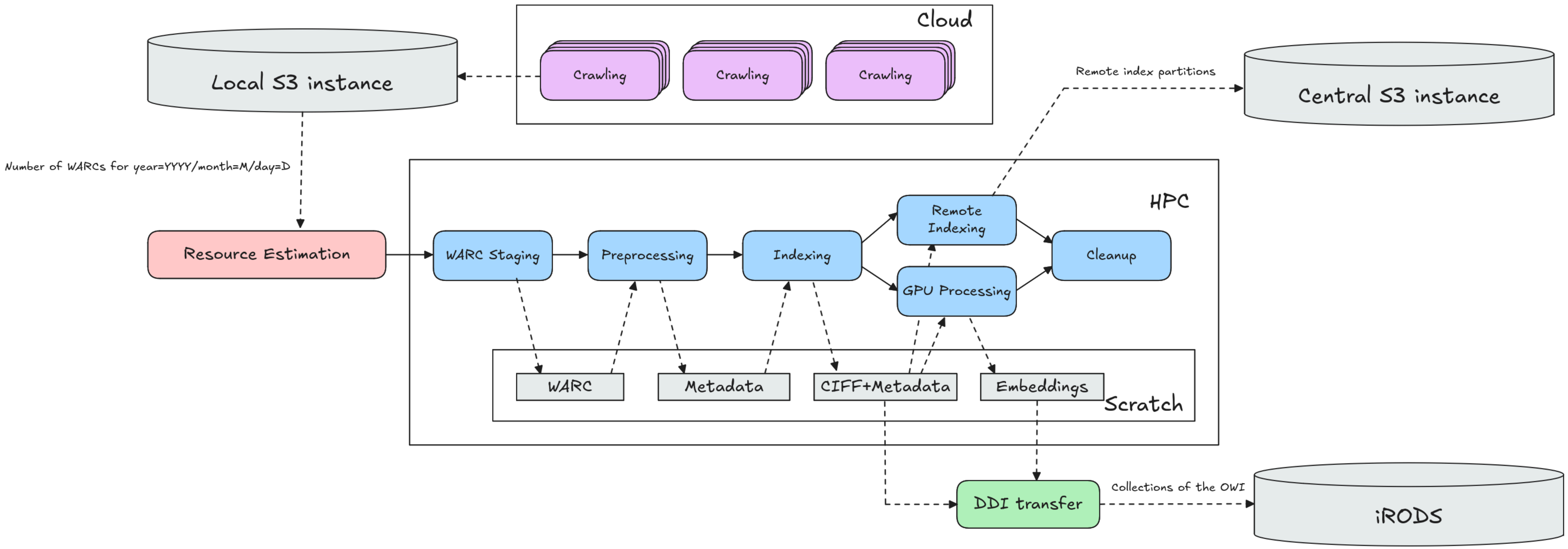
Rank	Language	Share
10	Dutch	1.5%

Building the Open Web Index

Conceptual Workflow



Daily Workflow Pipeline



Crawlig Infrastructure (2026)

- **25 VMs** across IT4I, LRZ, CSC, CERN
- **100 M URLs / day** crawling capacity
- **~1.1 PB** raw data · **~50 TB** index · **810+ public datasets**
- Federated storage: iRODS + S3, orchestrated via LEXIS

Limitations as of 2026

- TLR 5/6: it is not a product (yet)
- No 24/7 Operations
- Limited funding - minimum engineering / improvement

IN DETAIL

The OpenWebSearch.eu Pipeline — Overview

Step 1 — Crawling

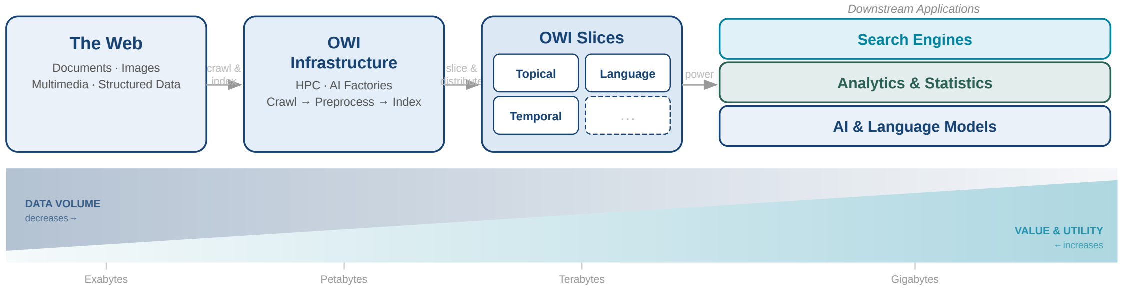
- 100 M URLs / day, 25 machines in 3–4 DCs
- European domain focus
- Respects robots.txt, Gen AI opt-out
- Full transparency via Dashboard

Step 2 — Preprocessing (Resilipipe)

- HTML → clean text (ratio 1:5)
- Metadata: JSON-LD, Microdata, OpenGraph
- Semantic enrichment: language, topics, NER, geo-locations
- Structured data from 60 % of pages

Step 3 — Index & Distribution

- Sparse full-text index (CIFF / Lucene)
- Feature index: link structure, style, genre
- Dense index: Jina V3 embeddings
- 810+ daily datasets via Owilix / Dashboard



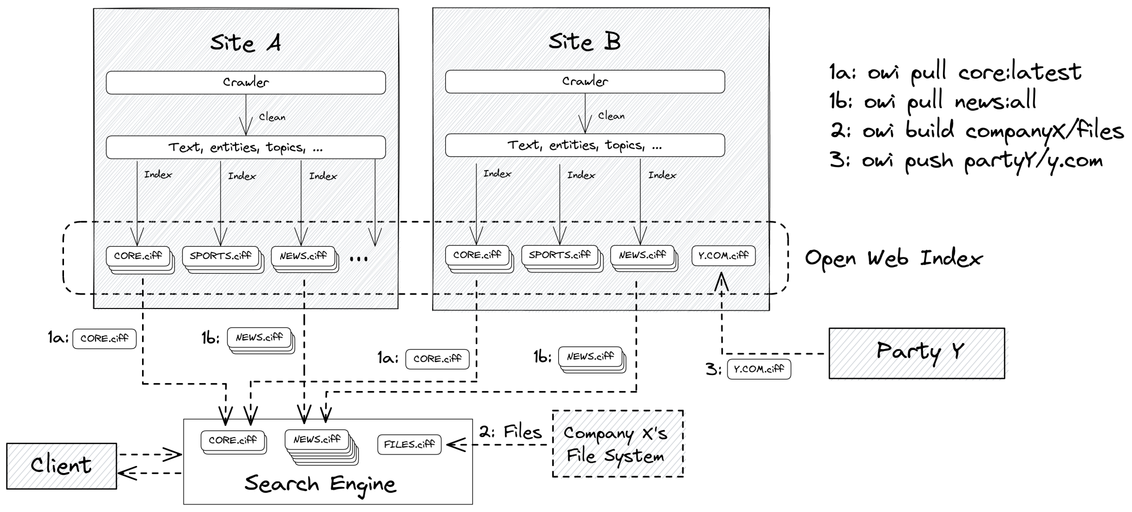
Scale (as of early 2026)

- **25 VMs** across 3–5 data centres (IT4I, CSC, LRZ, CERN) running 24/7
- **100 M URLs / day** crawling capacity
- **~1.1 PB** raw data · **~50 TB** index in **810+ public datasets**
- Factor **20 cheaper** than commercial indices

Architecture principles

- Federated storage (iRODS + S3), cross-DC workflow via LEXIS
- Client-side SQL access via Owilix CLI
- Open format: CIFF sparse index + Parquet files
- Dense embeddings (Jina V3) available on select slices

Architecture & Scale



Four Cluster Tiers per Data Center

- **CCT** — OWLer crawler (StormCrawler-based); produces WARC files into S3
- **CFT** — singleton ScyllaDB URL frontier at CSC; coordinates all crawl sites
- **PET** — Resilipipe (Python multiprocessing + Resiliparse): WARC → Parquet
- **IST** — Open Web Indexer (Apache Spark): Parquet → CIFF + remote index

Deployed at: IT4I@VSB · BADW-LRZ · CSC · CERN + FSTP partners

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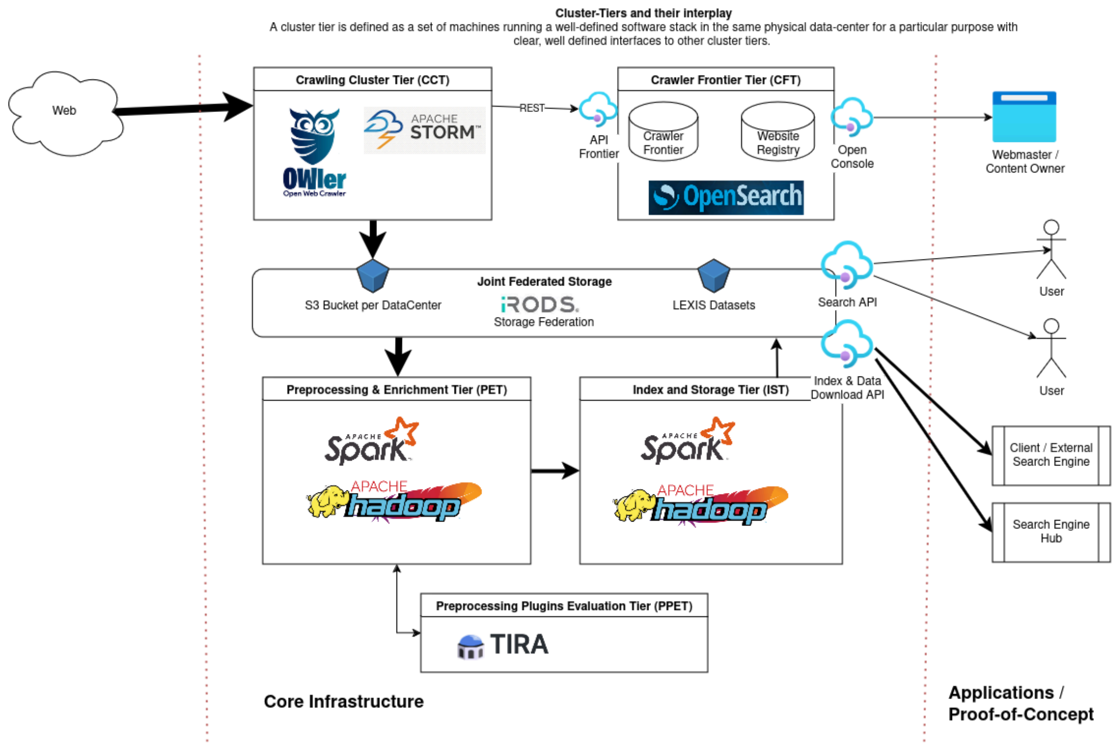
Cluster Tiers in Detail



Crawling (WP 1)

- **OWLer** (distributed StormCrawler): multi-site web crawl, WARC output
- Peak: ~100 M URLs/day at 4 partner sites (UNI PASSAU, LRZ, IT4I, CSC + EXAION)
- **ScyllaDB** frontier: dedup, politeness, cross-DC coordination

Preprocessing (WP 2)

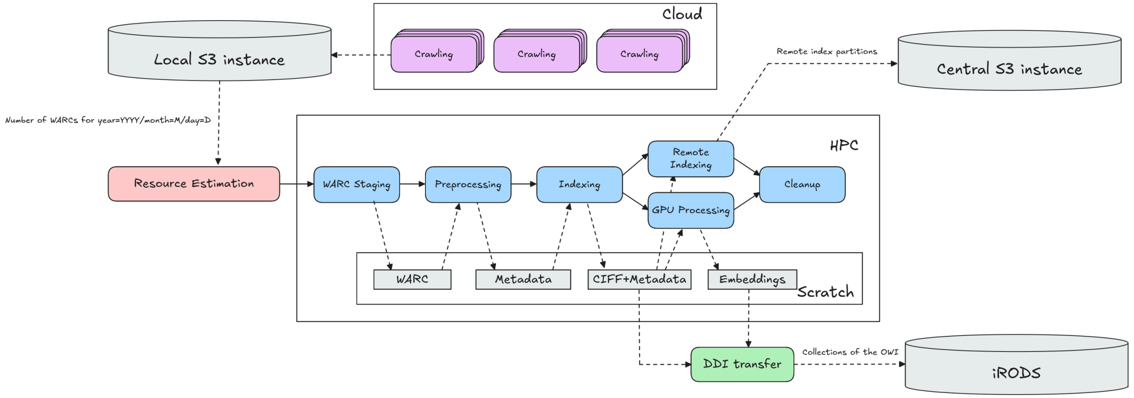
- **Resilipipe** on Spark: HTML → clean text, metadata extraction, topic/entity tagging
- **PPET** (singleton): plugin evaluation via TIRA before promotion to production
- GPU tier: quality-filtered docs → Jina V3 dense embeddings

Indexing (WP 3)

- **Open Web Indexer** on Spark: Parquet → CIFF shards per collection + language

- Also produces **remote index** (DuckDB/OUR²S-queryable Parquet partitions)
- Federated storage: **iRODS** + S3 via LEXIS DDI layer

Daily Workflow

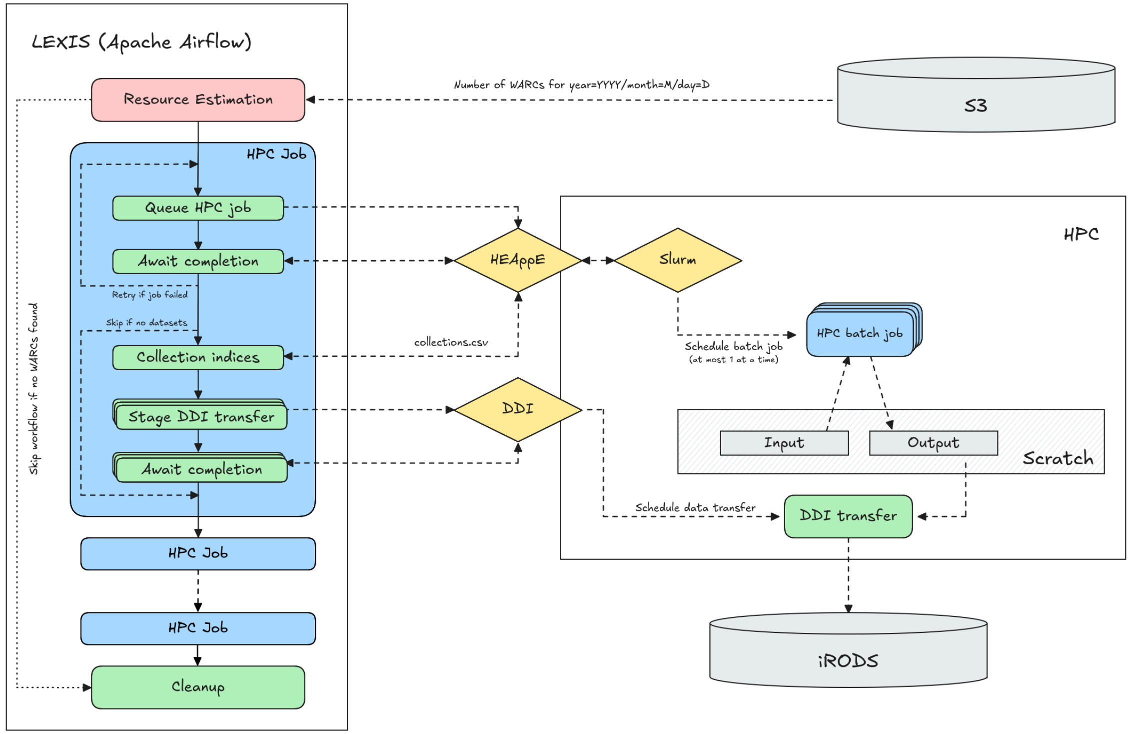


Each day at **03:00 AM** the pipeline triggers:

1. **WARC Staging** — copy new WARC files from S3 to HPC scratch
2. **Resilipipe** — WARC → gzipped Parquet (text, metadata, enrichments)
3. **Open Web Indexer** — Parquet → CIFF shards per collection / language
4. **GPU Processing** — Resilipipe GPU → Jina V3 embeddings for quality slice
5. **Remote Index** — Parquet partitions for DuckDB / OUR²S remote queries
6. **DDI Transfer** — push output collections to iRODS via LEXIS DDI layer
7. **Cleanup** — remove intermediate HPC scratch data

All intermediate data stored on **HPC scratch** between steps — restartable on partial failure.

Workflow Orchestration via LEXIS



LEXIS (Apache Airflow) manages the full daily pipeline:

- **Resource estimation**: counts WARCs for that date → computes Slurm allocation
- **HEAppE**: submits and monitors HPC batch jobs at each data center
- **DDI layer**: transfers output collections to iRODS after each job succeeds
- Automatic retry on failure; session tokens refreshed throughout

Workflow configuration (JSON per DC):

- `crawl_location` / `hpc_location` / `storage_location` — Airflow Connections
- Per-task: `resources`, `max_retries`, `stage_ddi`, `depends_on`
- Dev / test / prod variants with reduced resource profiles

Data center connections: S3 (crawl), HEAppE (HPC scheduling), iRODS (persistent storage) — fully configurable per data center.

Index & Federation

- **~1.1 PB** raw data · **~50 TB** index data · **810+ public datasets**
- **10–20 TB** new index data per month
- Approx. **factor 20** smaller than commercial indices (factor 10–50× for multimedia)

Federated Architecture

- Distributed over **IT4I, CSC, LRZ, CERN** — onboarding further AI Factories
- Cross-DC workflow execution via LEXIS platform
- Federated storage: iRODS + S3; client-side SQL access via owilix / DuckDB
- Dense embeddings available on select slices

Step 1: Crawling the Web

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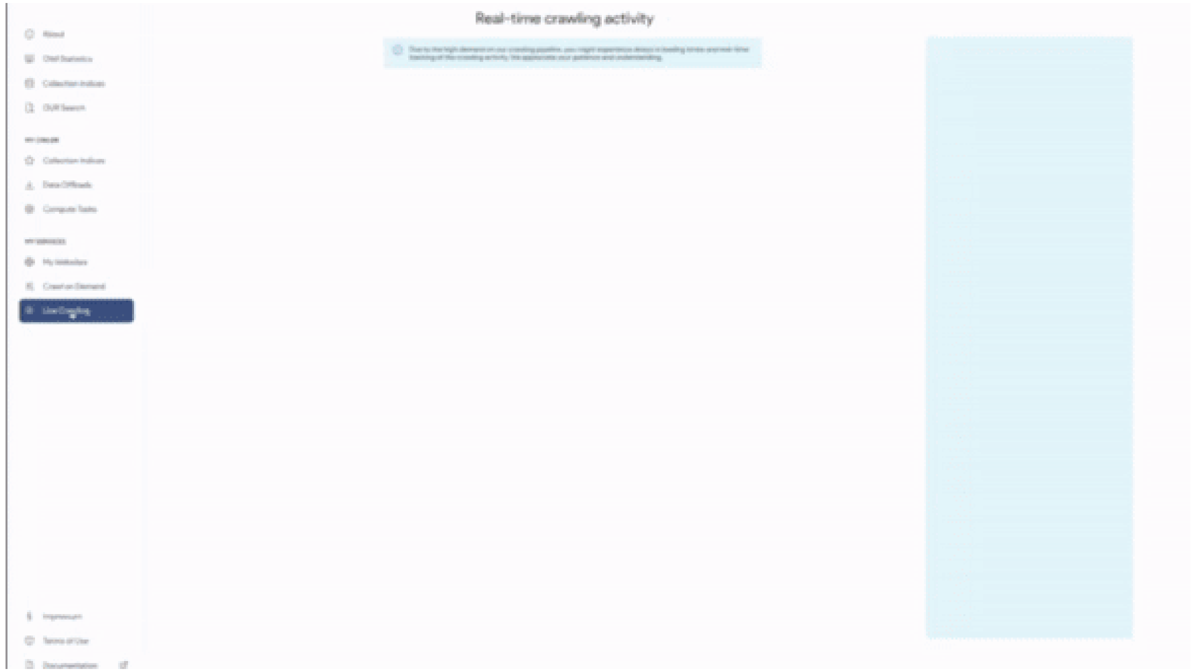


OWLer — Open Web Crawler

- Up to **105 M URLs / day** across 4 European data centers
- Central URL Frontier at **CERN** — 2 B+ URLs in crawl space
- **1.25 PiB** collected · **1,927 machine-days** · ~39.6 B fetches
- European language and domain bias
- Only HTML; No headless browsing

Constraints

- Downstream processing caps: **1–2 TB/day per DC**
- Factor ~20× below commercial freshness
- Requires 2–4 dedicated engineers for 24/7 ops

IN DETAIL

OWLer — Three-Tier Crawling Architecture

Tier 1 — Crawling Cluster (StormCrawler)

- StormCrawler on Apache Storm — distributed, fault-tolerant, linearly scalable
- VMs at **LRZ**, **IT4I**, **CSC** (OpenStack)
- Three pipeline modes:
- **Regular** — explorative hyperlink-following (highest coverage, highest risk)
- **Sitemap** — sitemap-driven discovery (lower risk, curated scope)
- **WARC2WARC** — ingest external WARCs (e.g. Common Crawl) without live fetching
- Outputs WARC files → fed into downstream preprocessing pipeline

Tier 2 — URL Frontier (CERN)

- Central gRPC service coordinating all crawlers
- Manages 2 B+ URL crawl space in OpenSearch
- Stream-based and batch-based APIs
- Politeness: delay between fetches per domain

Tier 3 — Persistence

- OpenSearch backend for URL state and logs
- Apache Flink for real-time log aggregation
- Published as OWI Dashboard datasets

URL Frontier — Coordination & Scheduling

Centralized URL Management

- Single authoritative component — simplifies monitoring, debugging, coordination
- gRPC API with Protocol Buffers — language- and crawler-neutral
- Benchmark: **295.9 M simulated fetches in 12 h**, 2 B URL crawl space, 80 GB RAM peak
- Up to **5,500 URLs/second** (batch mode), 2,000/s (stream mode)

Scheduling Strategy

- Each URL gets a `nextFetchDate` — continuous, not round-based
- Freshness via `If-Modified-Since` + MD5 content hash comparison
- **Quality lists** to prioritize high-value content:
  - ClueWeb22 Super Head (Category B)
  - Curlie / DMOZ directory (human-curated)
  - Wikipedia external link outlinks

URL Tagging *(% of crawl space)*

Tag	%
HTML	83.1 %
GenAI allowed	79.6 %
Index allowed	77.0 %
Wikipedia outlink	48.7 %
SuperHead ClueWeb	9.66 %
Curlie category	~3.7 %
Blacklisted	0.18 %

Crawl Statistics — Three Operational Phases

Three major crawl campaigns across 120 weeks (Nov 2023 – Feb 2026) at European HPC centres

Phase	Machine-days	Data (TiB)	Fetches (B)	Hosts (M)	Unique pages	Recrawl ratio
Aug 2023 – Aug 2024	760	315	10.76	40.5	1.17 B	9.2×
Apr 2025 – Jul 2025	189	175	4.76	162.0	3.76 B	1.2×
Dec 2025 (validation)	—	—	—	20.5	341 M	—
Total	**1,927**	**~1,280**	**~39.6**	—	**~12.6 B**	—

Peak throughput

~105 M URLs/day
88% fetch success rate
184 languages detected

Practical cap

1–2 TB/day/DC
(downstream processing limit)
~20× below commercial freshness

To reach production scale

2–4 dedicated engineers
24/7 continuous operation
~50× more bandwidth + storage

Auxiliary Crawls — Targeted Fill-In Collections

Beyond general-purpose web crawling, specialized scrapers harvest structured sources that standard hyperlink-following cannot reach:

Source	Coverage	Volume
Wikipedia	10 languages (en, de, fr, es, it, nl, pl, sv, ar, ja)	83 GB WARC/month · 6.5 M external domains extracted
Mastodon	150 instances (top 1,000 by activity)	5 GB/day · 150 GB/month
OpenAlex	Scientific publications snapshot + incremental API	117 GB compressed WARC
Pangaea / OpenAire / Eudat	Scientific data repositories	via API → WARC
Internal GitLab	Code repositories	via API → WARC

All auxiliary crawlers produce **WARC files** and feed into the same downstream OWI pipeline as the general-purpose crawler.

Why auxiliary crawls?

- Wikipedia outlinks seed high-quality URLs into the general-purpose frontier
- Scientific publications enable Open Science Search use cases
- Mastodon adds social-web content beyond the crawlable public web
- Complements gaps in general-purpose coverage (API-only content, structured dumps)

Compliance, Transparency & Webmaster Control

Ethical Crawling Practices

- Respects **robots.txt**, `X-Robots-Tag`, `noindex` directives
- Two derived binary flags stored in every Parquet record:
  - `Index` — allowed for web indexing / search
  - `GenAI` — allowed for generative AI model training

Index	GenAI	Share
✓	✓	**76.2 %**
✓	✗	0.74 %
✗	✓	3.34 %
✗	✗	19.7 %

- **Demarcated crawling subnets** — IP ranges separated from institutional access to avoid license infringements
- **Blacklisting**: Ultimate Hosts Blacklist + Université Toulouse Capitole lists (0.18% of URLs blacklisted)

Open Webmaster Console

- Any webmaster can look up their domain in the OWI
- **Domain verification** to unlock management
- **Takedown requests** at URL or subdomain level → propagated to all index users
- **Crawl constraint config** via OWLer wizard
- Ethical self-assessment survey required before dataset download

Incident log: Bank of America (robots.txt ambiguity), ACM Digital Library (licensed content), anonymous platform (forum spam false positive) — all investigated and resolved

Step 2: Preprocessing

Step 2: Preprocessing

Resilipipe — WARC → Parquet Pipeline

- Two-stage pipeline: **Resilipipe (CPU)** for extraction & enrichment, **Resilipipe GPU** for embeddings & classifiers
- Modular architecture — 10+ pluggable modules (links, structured data, quality scoring, trigger warnings, …)
- Custom parallelization on HPC: SLURM + Apache Airflow orchestration
- Output: column-oriented **Apache Parquet** with per-document metadata, embeddings, and prediction scores

Scale & Constraints

- **108 M documents** embedded with jina-embeddings-v3 (785 GB across 55 published datasets)
- GPU stage bottleneck: embedding throughput limits end-to-end pipeline speed
- Downstream cap of **1–2 TB/day per data center** inherited from crawling

Key Enrichments

- Content extraction & language detection
- Link graph & structured data (FAQs, geo, licenses)
- Curlie taxonomy labels
- Quality scoring (ONNX)
- Trigger warnings (14 languages, 19 categories)
- Dense embeddings (jina-embeddings-v3)
- Phishing classifier & LLM detector

IN DETAIL

Resilipipe (CPU) — Modular WARC → Parquet Pipeline

Architecture

- Input: **WARC files** from OWLer crawl → Output: **Apache Parquet** (columnar, query-friendly)
- Custom parallelization replacing PySpark:
- **ProcessWorkers** → **ParsingThreads** → **WritingThread**
- Three module types:
- **BasicModule** — operates on raw WARC records (headers, links)
- **HTMLModule** — receives parsed HTML DOM tree (structured data, quality)
- **PostProcessingModule** — runs after all others (collection indices, forum classification)

CPU Modules

Module	Function
`ows_headers`	HTTP headers & metadata
`links`	Link graph extraction
`curlielabels`	Curlie taxonomy classification
`structured_data`	FAQs, licenses, geo, opening hours, phone
`pyterrier_quality`	Quality scoring (ONNX)
`trigger_warnings`	Content warnings (14 langs × 19 types)
`collection_indices`	Per-collection index metadata

Module	Function
`forum_classification`	Forum / discussion detection

Resilipipe GPU — Embeddings & Prediction Heads

Embedding Pipeline

- Model: **jina-embeddings-v3** — multilingual, 8192-token context, task-specific LoRA adapters
- Selected from evaluation on 3.3 GB multilingual OWI test dataset
- Chunking via embedding model's tokenizer — embeddings computed **per chunk** (supports RAG retrieval)
- **55 published datasets** containing embeddings for **108 M documents** (785 GB total)
- Research track: embedding compression to reduce storage footprint

GPU Parallelization

- ReadingWorker → EmbeddingWorker (per GPU) → AggregationWorker → ModuleWorker (per GPU) → WritingWorker
- Prediction heads: small `torch.nn.Module` networks trained on chunked embeddings
- Each GPU processes batches independently; aggregation worker merges results

Prediction Heads

Head	Task
Phishing classifier	Detect phishing / scam pages
LLM detector	Flag LLM-generated content

Why jina-embeddings-v3?

- Best multilingual performance on OWI evaluation
- Task-specific adapters (retrieval, classification, clustering)
- Long-context support (8192 tokens)
- Open-weight model

Computational Ethics — Content Safety & Detection

Trigger Warning System

- Lexical scoring approach across **14 languages** and **19 warning categories**
- Categories include: violence, self-harm, sexual content, hate speech, substance abuse, eating disorders, and more
- Per-document scores stored in Parquet output — downstream consumers can filter at chosen thresholds
- Designed for transparency: users of OWI data can apply their own policies

LLM-Generated Content Detection

- Native advertising detection research (CLEF 2025 shared task)
- LLM detector prediction head identifies machine-generated text
- Critical for data quality: prevents AI-generated content from polluting training corpora and search indices

Why this matters

- Web-scale data inevitably contains harmful content
- Index consumers (search engines, researchers, AI trainers) need **per-document signals** to make informed filtering decisions
- OWI provides the signals — **not** the filtering policy

Design Principle

Signal, don't censor — OWI enriches data with safety metadata; downstream users decide thresholds and policies.

HPC Deployment — SLURM + Apache Airflow

Orchestration Stack

- **Apache Airflow** manages DAG workflows across both Resilipipe stages
- **SLURM** scheduler allocates CPU and GPU nodes at HPC data centers
- Automated WARC staging: **S3** → **scratch filesystem** before processing

Workflow Stages

1. **WARC Staging** — pull WARC files from S3/iRoDS to local scratch
2. **Resilipipe (CPU)** — extract, enrich, write Parquet
3. **Resilipipe GPU** — compute embeddings, run prediction heads, write enriched Parquet
4. **Publication** — upload enriched datasets to OWI distribution (LEXIS / S3)

Deployment at Scale

- Runs at **LRZ**, **IT4I**, **CSC** HPC centres
- CPU and GPU jobs submitted as SLURM batch arrays
- Airflow DAGs monitor progress, handle retries, track dataset lineage

End-to-End Flow

```
WARC (crawl)
  → Resilipipe CPU
    → Parquet (enriched)
      → Resilipipe GPU
        → Parquet (+ embeddings)
          → OWI distribution
```

Step 3: Building the Index

Index Format: CIFF + Parquet

CIFF (Common Index File Format)

A Protobuf schema for sparse inverted files — the OWI standard interchange format:

- **Header** — collection stats: document count, unique terms, avg. document length
- **Postings lists** — per term: document frequency + collection frequency + (docid, tf) pairs
- **Document records** — internal numeric ID ↔ external URL + document length

Designed so any existing search engine can import the data and start ranking immediately.

Parquet Metadata Files

Distributed alongside each CIFF shard — aligned partitions containing:

- URL, title, language, crawl date
- Quality / importance scores from Resilipipe
- Topic / entity enrichments (WP 2)
- Enables SQL-style analytics without downloading the full index

Dense Embeddings (GPU slice)

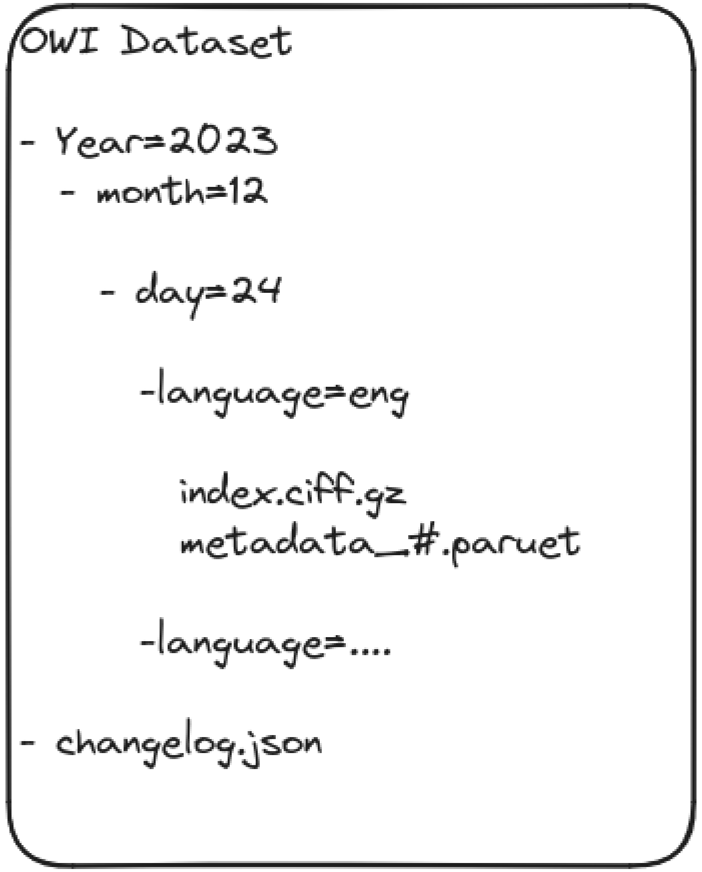
- High-quality / high-importance doc subset
- Jina V3 embeddings, 884 GB in public datasets
- Enables semantic search and RAG on OWI content

Dataset Structure & Versioning

Each **daily shard** follows a consistent folder hierarchy:

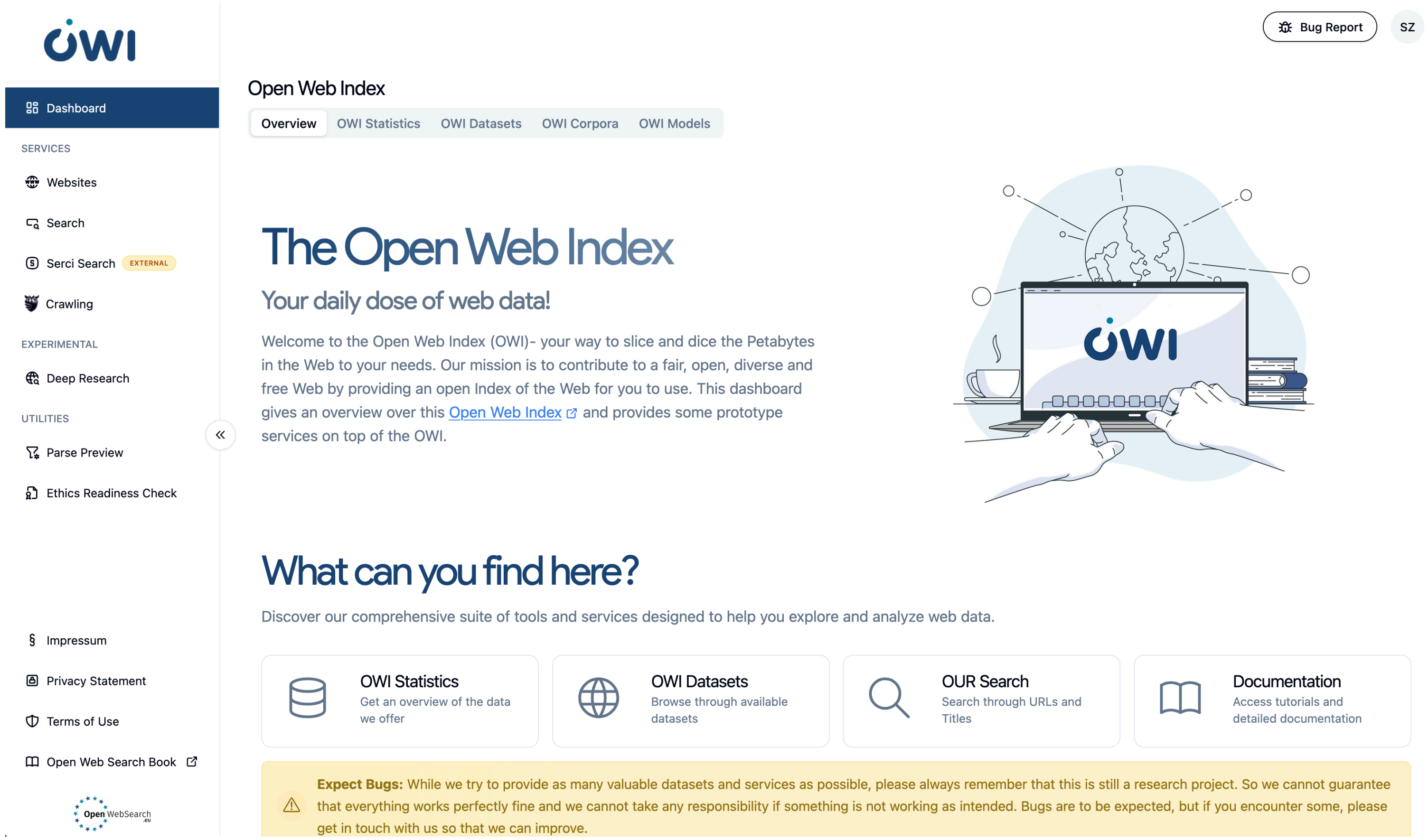
```
OWI Dataset
  └── year=YYYY
        └── month=MM
              └── day=DD
                    ├── language=eng
                    │     ├── index.ciff.gz
                    │     └── metadata_*.parquet
                    ├── language=deu
                    │     └── ...
                    └── changelog.json
```

- One dataset per data center per day
- Collections: `main`, `curlie_full`, `gpu` (embeddings), and domain-specific slices
- `changelog.json` tracks document additions and deletions (take-down compliance)
- Git-like versioning via **owilix**: `pull`, `push`, `build`



Accessing the Open Web Index

The Dashboard — openwebindex.eu

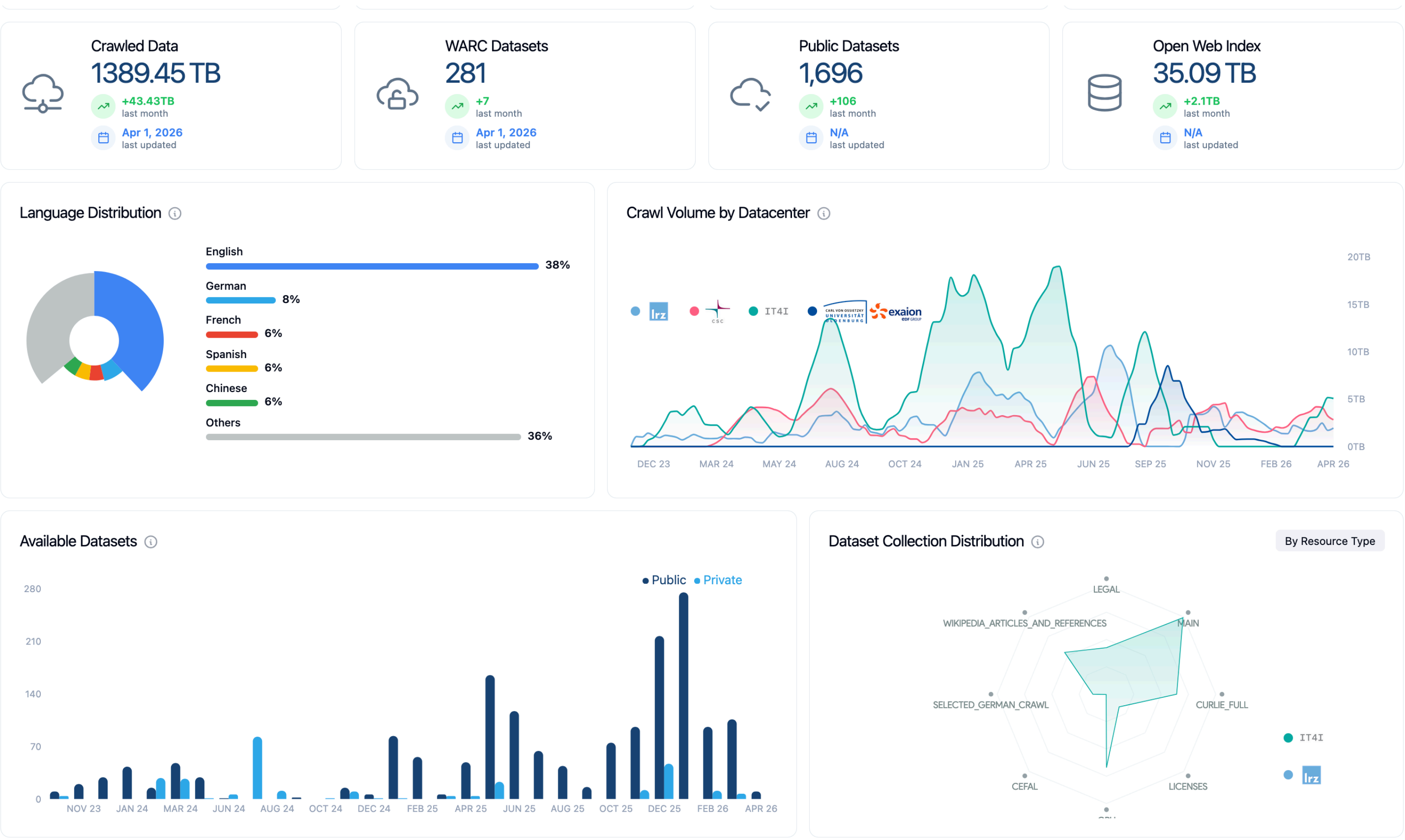


424 registered users · 17,000+ unique visitors · 70,000+ page views

Central entry point for the Open Web Index openwebindex.eu

- **OWI Statistics** — crawl volume, index size, language distribution, dataset timeline
- **OWI Datasets** — browse, filter and download index shards (by DC, date, format, language)
- **Search** — demo content search + SERCI integration

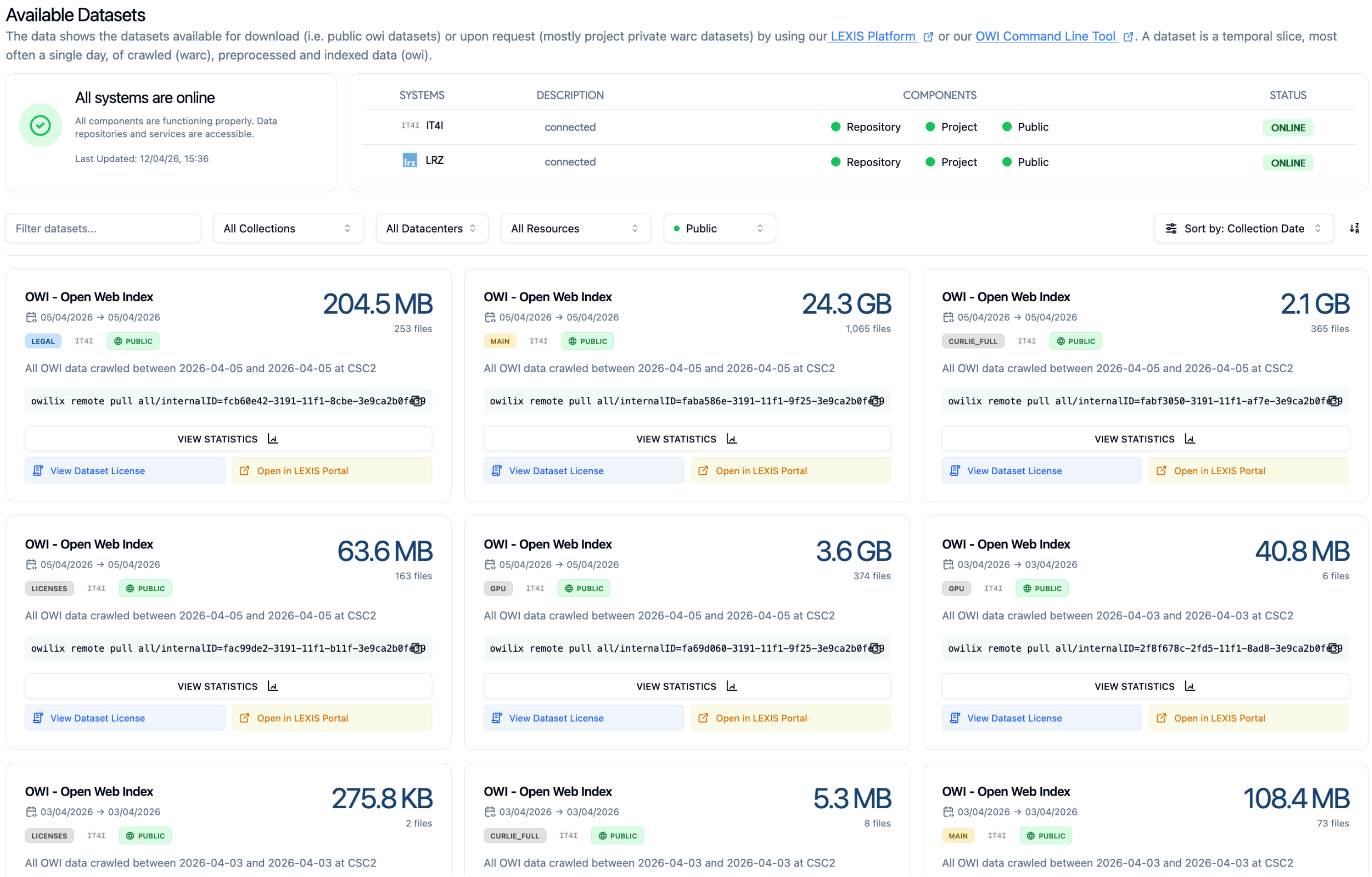
Dashboard — Statistics & Datasets



OWI Statistics (04/26)

- **1,389 TB** crawled · **281** WARC datasets · **1,696** public datasets · **35.09 TB** index
- Language distribution chart (English 38%, German 8%, French 6%, ...)
- Crawl volume over time per data center + dataset collection distribution by type
- Datasets over time

Dashboard — Statistics & Datasets



OWI Datasets Tab

- Status of repositories
- Filter by datacenter, format (WARC / Parquet / CIFF / Embeddings), language, date
- Dataset cards with download links — LEXIS Portal or owilix CLI
- Parquet file preview directly in the browser

Download Tool: owilix

owilix — The OWI Command-Line Tool

Git-like versioning for index datasets — pull, build, push:

```
owilix remote pull all:latest/collectionName=main
owilix local ls all:latest
```

Slice by: **domain**, **language**, **topic** (Curlie), **sitelist**, **structured data**, **outlinks**

Authentication via B2ACCESS / Keycloak (LEXIS SSO)

OWILIX - Open Web Index CLI

version 5.3.1 python 3.11+ license Apache 2.0

A command-line tool for managing OpenWebIndex datasets across local and remote repositories.

Quick Start

One-Line Install (Recommended)

```
curl -fsSL https://openwebindex.net/owilix/install.sh | sh
```

https://opencode.it4i.eu/openwebsearcheu-public/owi-cli/

Remote Index — Query first, then Download

- Parquet partitions directly queryable via **DuckDB** or **Apache Spark**
- Full-text search
- SQL access: filter by language, domain, date — no full-shard download needed
- Enables search + filtering + download (but still in alpha phase)

```
~ owilix remote search "michael granitzer" --limit 10
Searching remote index: terms=['michael', 'granitzer'] language=eng representation=main_content mode=OR limit=10
Found 10 results in 113.4s
                    Search results (10 results in 113.4s)
```

score	date	url
3.088	2025-12-30	https://easychair.org/smart-program/ICADL2024/person16.html
3.010	2026-02-19	https://www.wolframalpha.com/input/?i=michael
3.010	2026-01-19	https://www.michaelmoennich.com/
3.010	2025-12-08	https://michaelkoenig.de/
3.010	2026-01-05	http://www.michaelmoerk.dk/
3.010	2026-01-22	https://michaelkoenig.de/
3.010	2026-02-18	https://www.sktthemes.org/forums/users/dedicationpt/replies/
3.010	2026-02-26	https://www.abeldiaz3.com/tag/michael/
3.010	2026-02-20	https://discourse.psychopy.org/u/michael
3.010	2026-02-27	https://thedigitalgobag.com/author/michael/

The OpenWebSearch Book

Comprehensive Documentation at openwebindex.eu/book

Covers the full OWI ecosystem:

- Getting started with owilix
- Building vertical search engines with MOSAIC
- Data formats: CIFF, Parquet, embeddings
- Pipeline architecture and HPC deployment
- Legal & ethical guidelines for index users

Published as an open, continuously updated online resource — aligned with deliverables and community feedback.

Access Modes at a Glance

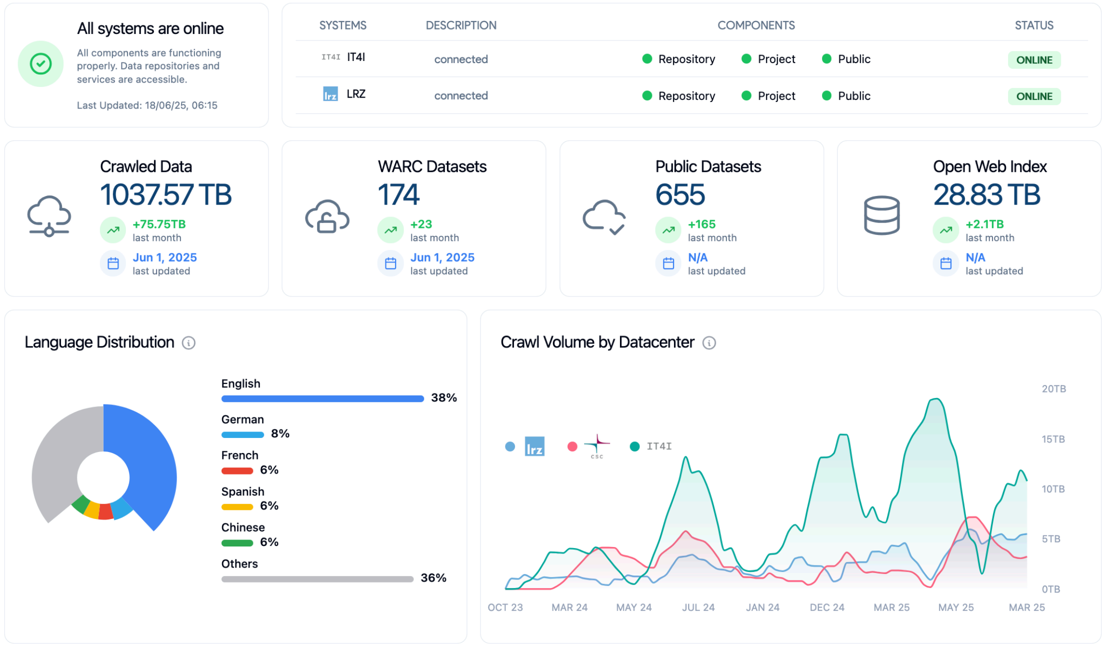
Tool	Mode	Best for
Dashboard	Browser	Explore, monitor
owilix	CLI	Download, slice, build
Remote index	DuckDB/SQL	Query + download
MosaicRAG	Conversational	RAG & chat over OWI

All software, data formats and access tools are **open source**

IN DETAIL

Accessing the Open Web Index

Dashboard — Statistics & Datasets in Detail



OWI Statistics Tab

- Repository health per data center (IT4I · LRZ)
- Key indicators: pipeline runs, HPC core-hours consumed, avg. duration, active pipelines
- Crawled data (1.3 PB), WARC datasets, public datasets, index size
- Language distribution chart + crawl volume timeline per DC
- Dataset collection distribution by resource type (main / GPU / curlie / embeddings)

OWI Datasets Tab

- Filter: datacenter · format (WARC / Parquet / CIFF / Embeddings) · language · date range
- Infinite-scroll dataset cards: title, date, DC, public/private badge, file listing
- Parquet file preview in-browser; download via LEXIS Portal or `owilix remote pull`

Dashboard — Corpora, Models & Search

OWI Corpora

Curated, administrator-prepared collections for distribution:

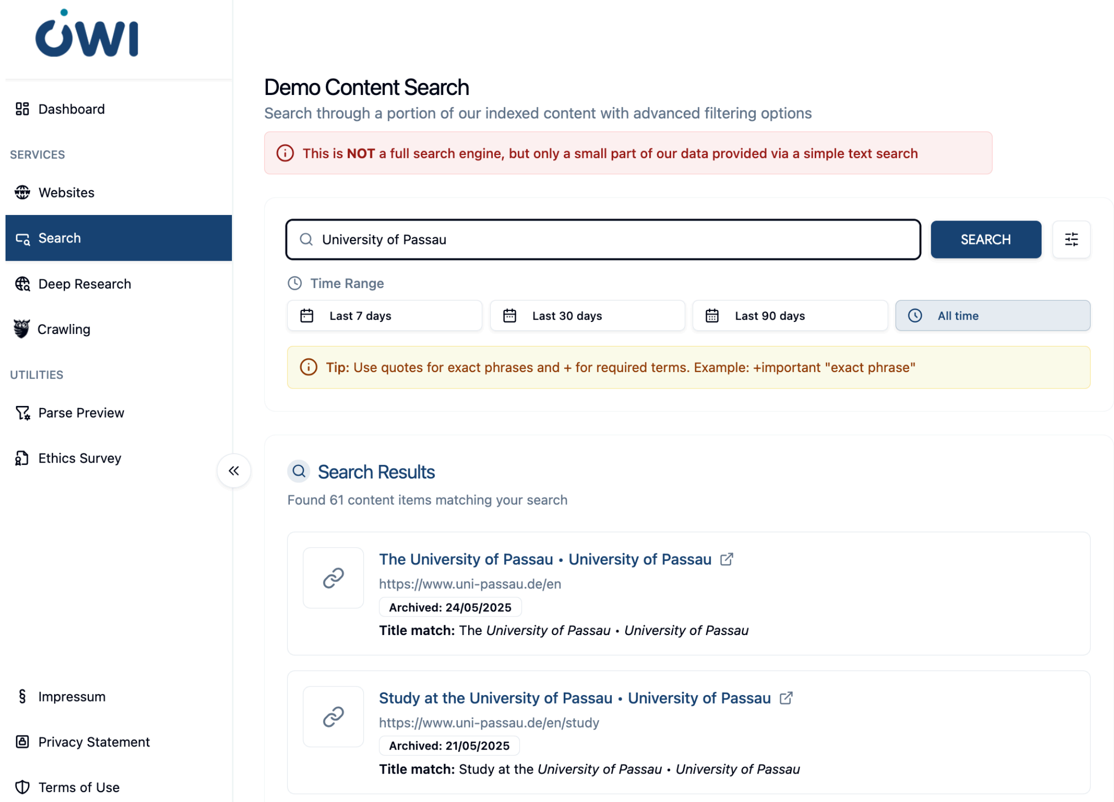
- Licensing metadata per corpus (creator, terms, full licence text)
- LEXIS integration for storage, access control and metadata
- Includes domain-specific and project-specific collections

OWI Models

Registry of language models for web search:

- Browser-runnable models via WebAssembly (Wllama library)
- Cloud-based models (OpenAI, DeepSeek) for heavy tasks
- Model cards with training data, benchmarks, and GGUF file listing

OUR Search (Demo)

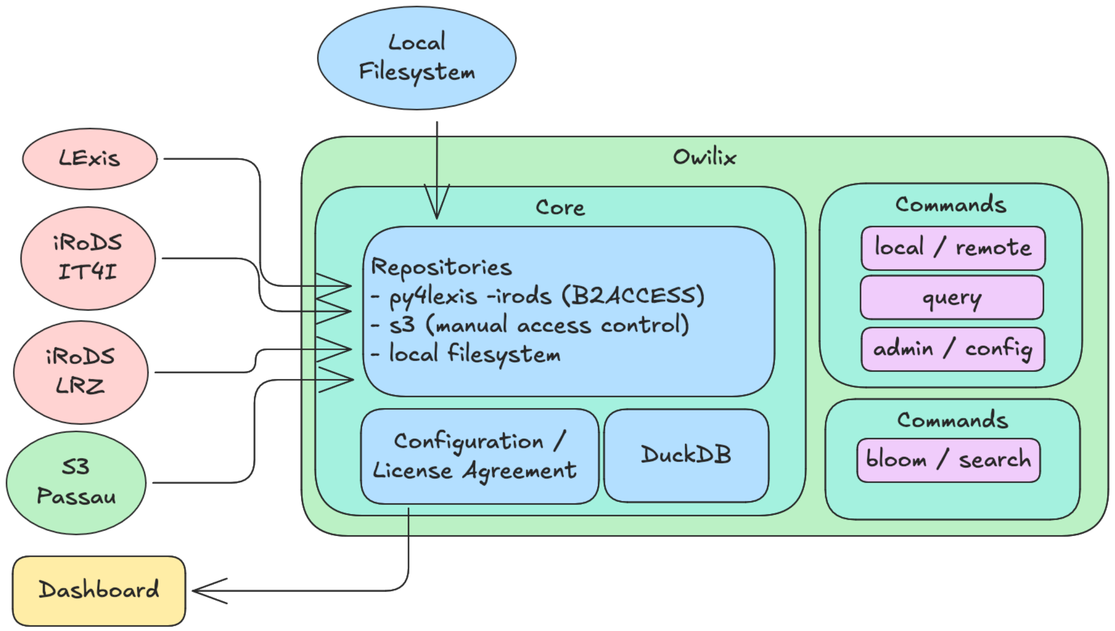


Simple + advanced field-specific search over a subset of the index; time-range filtering; result cards with thumbnail, title, URL

Deep Research (Experimental)

AI-powered multi-step research: configurable depth + breadth, local or cloud LLM, interactive clarification mode

owilix — The Open Web Index Client



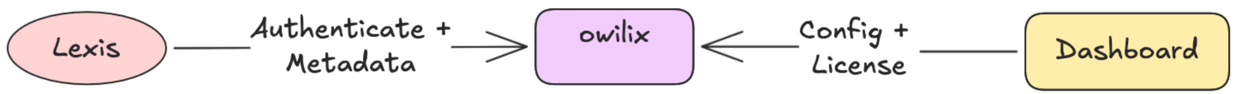
Four main workflows:

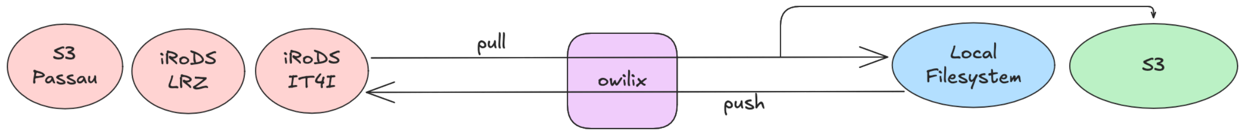
1. **Authentication** — B2ACCESS via LEXIS + Dashboard config (OpenID Connect / Keycloak)
2. **Sync Datasets** — pull/push between federated iRODS/S3 storage and local filesystem
3. **Query / Slice** — client-side SQL queries with JSON, HTTP, OpenSearch output
4. **WARC Access** — fetch original WARC files via SQL querying

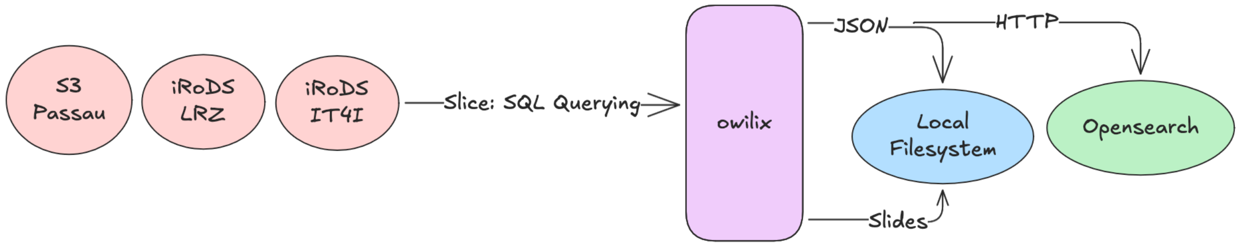
Slice the index by:

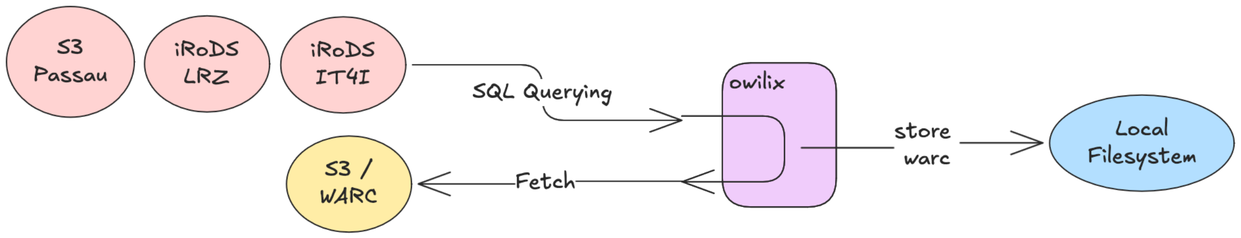
- **Domain** (.de, .at, .fr, …)
- **Topic** (Curlie.org hierarchy labels)
- **Language** (filename-based selection)
- **Sitelist** (CSV of URLs / domains / TLDs)
- **Structured data** / microdata type
- **Outlinks** for link graph analysis

owilix Workflows









Remote Index — Query Without Downloading

What is the remote index?

A DuckDB / Apache Spark-queryable representation of the OWI — Parquet partitions aligned with CIFF shards, accessible from remote clients without downloading the full index.

Example queries

```
-- Count documents per language
SELECT language, COUNT(*) as docs
FROM read_parquet('remote://lrz:latest/main/**')
GROUP BY language ORDER BY docs DESC;

-- Filter by domain
SELECT url, title FROM ... WHERE domain = 'wikipedia.org';
```

Use cases

- Analytics and statistics on the live index
- Lightweight RAG pipeline ingestion
- Ad-hoc exploration before committing to a full shard download

OUR²S — Online Search API

OUR²S (built on Vespa) extends the remote index with a full REST search API:

- Structured document model: full page · snippets · embeddings
- Keyword, semantic and hybrid ranking modes
- Multi-index federation across data centers
- No local index needed — query directly over network

> → OUR²S details in the dedicated OUR²S column

Access summary

Mode	Tool	Authentication
Browse	Dashboard	Anonymous / B2ACCESS

Mode	Tool	Authentication
Download	owilix	B2ACCESS (LEXIS)
SQL queries	DuckDB remote	B2ACCESS
REST search	OUR²S API	API key
Conversational RAG	MosaicRAG	Open / B2ACCESS

MosaicRAG — Building RAG Systems on OWI

From OWI Datasets to Conversational Search

MosaicRAG consumes OWI daily datasets directly — CIFF inverted index files, Parquet metadata records, and dense embeddings (published since Dec 2025) — and exposes them through a modular conversational interface.

Pipeline for building your own MosaicRAG instance:

1. Download OWI slice with `owilix` (topic, domain, or language filter)
2. Ingest CIFF → inverted index; Parquet → DuckDB; embeddings → ChromaDB
3. Configure retrieval pipeline: sparse, dense, or hybrid
4. Choose reranker (TF-IDF · Embedding cross-encoder · Tournament LLM · Group LLM)
5. Optionally add summarization and relevance marking

> Full documentation: openwebindex.eu/book — "Building vertical search engines with MOSAIC"

Supported LLMs

Task	Models
Reranking	gemma2, qwen2.5, llama3.1
Summarization	+ DeepSeekv3
Relevance marking	gemma2, qwen2.5, llama3.1

Live Demo

Try conversational search on the Open Web Index at

mosaicrag.ows.eu

Using the Open Web Index — Applications

Using the Open Web Index — Applications

Realised Applications

- **Mosaic & MosaicRAG** — modular vertical search framework; science, health, arts demos → mosaic.ows.eu
- **Earth Observation Search** — RAG QA + geo-contextualised search (DLR / Radboud)
- **Multilingual Scientific Index** — enriching Finland's Research.fi portal (CSC)
- **Restaurant Recommender** — privacy-preserving mobile app; 214-user pilot (A1 Slovenia)
- **Argumentation Search** — args.me extended with OWI pro/contra index → args.me/owi

Community Applications (FSTPs)

- **AKSE** — argumentation knowledge graphs
- **VERITAS / DEXAI** — verification-oriented RAG
- **OMMS** — enriching OpenStreetMap with POI metadata
- **DTCommerce** — SME e-commerce product enrichment
- **CIFFIL** — cross-collection index for municipal search
- **TILDE** — trustworthy health domain search

The OWI enables a full ecosystem of vertical search engines, RAG systems, and domain-specific applications

IN DETAIL

MosaicRAG — Conversational Search on the OWI

Modular RAG framework extending MOSAIC with LLM-based capabilities and dense retrieval (WEBIS / TU Graz)

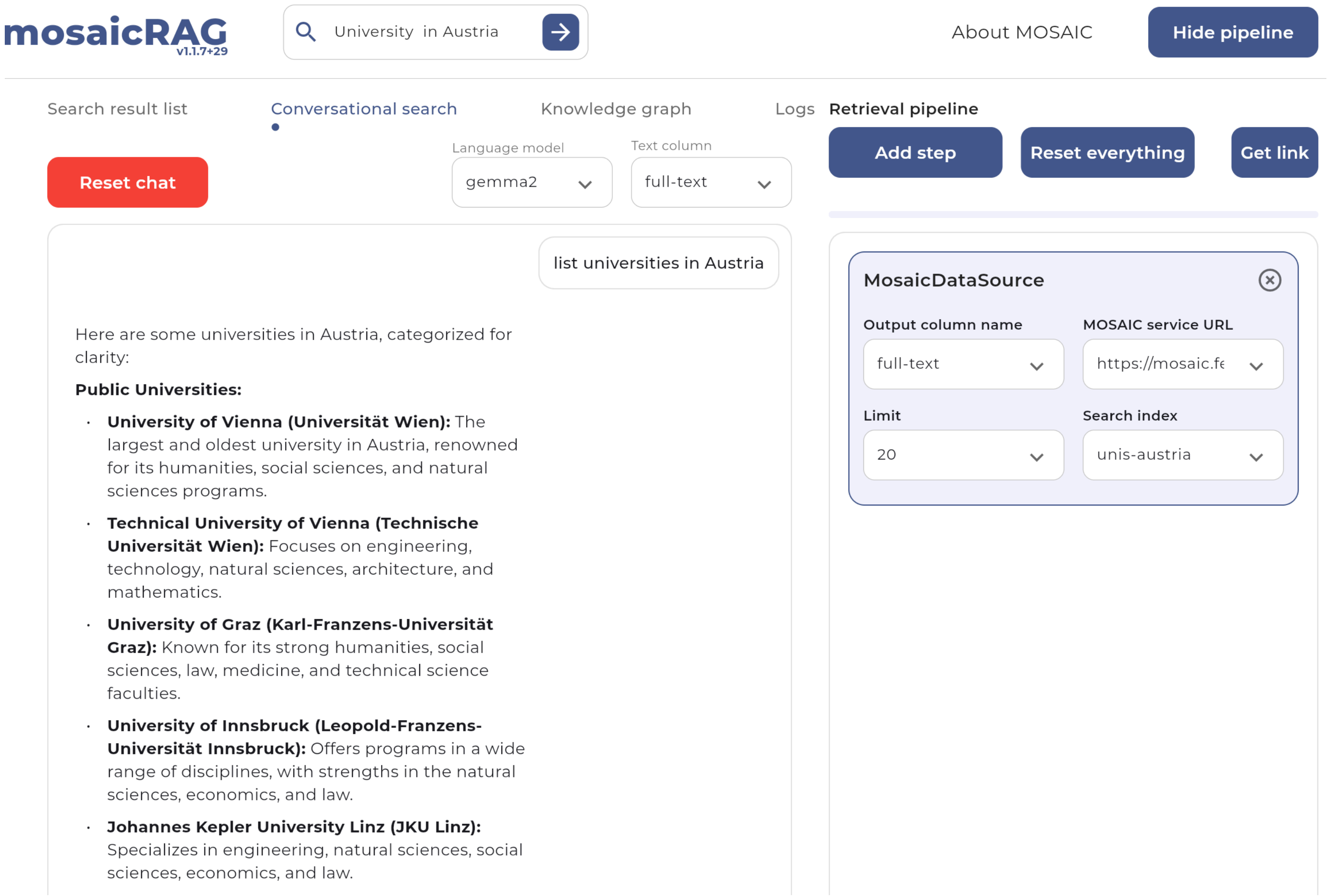
- **Two retrieval paths:** inverted index (CIFF) + dense retrieval
- **4 reranking strategies:** TF-IDF, Embedding cross-encoder, Tournament-style (LLM pairwise), Group-style (LLM window)
- **LLM backends:** self-hosted on-premise models for summarization
- **Conversational search:** follow-up questions about retrieval results; dual-window UI (search results + chat)
- **Workflow:** define retrieval pipeline → issue query → converse about results

Key Features

- Relevance marking — LLM highlights relevant passages within documents
- Summarization — user-provided instructions applied per document
- Domain-specific indices: science, health, arts (8M+ docs)
- Fully open-source modular pipeline

Data Pipeline

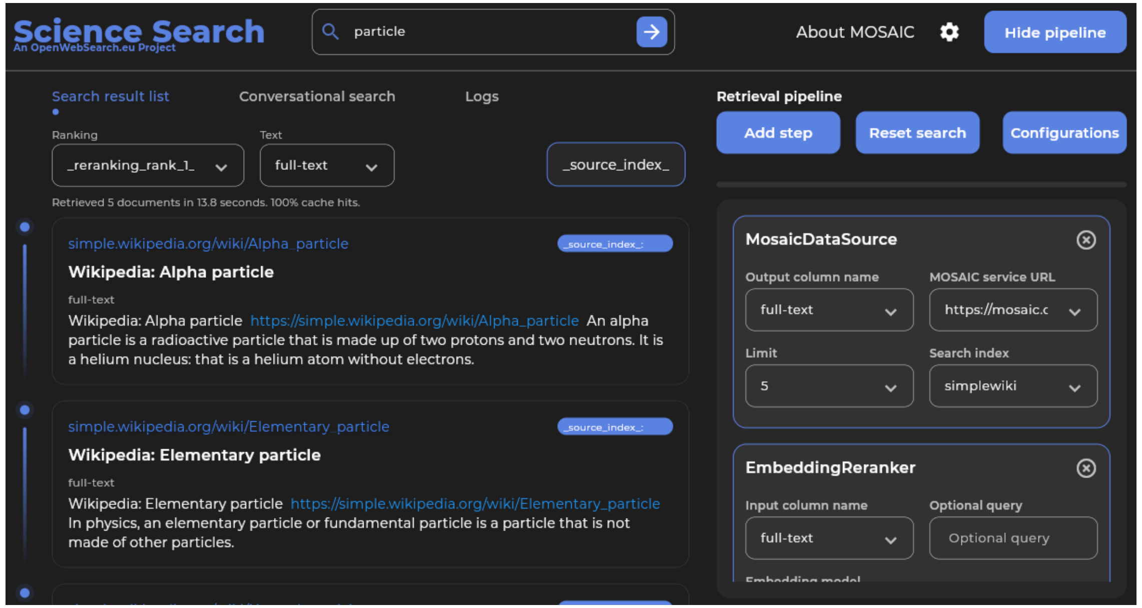
OWI daily datasets → CIFF (inverted index) + Parquet (metadata) + dense embeddings (since Dec 2025) → MosaicRAG



Demo: mosaicrag.ows.eu · mosaic.ows.eu

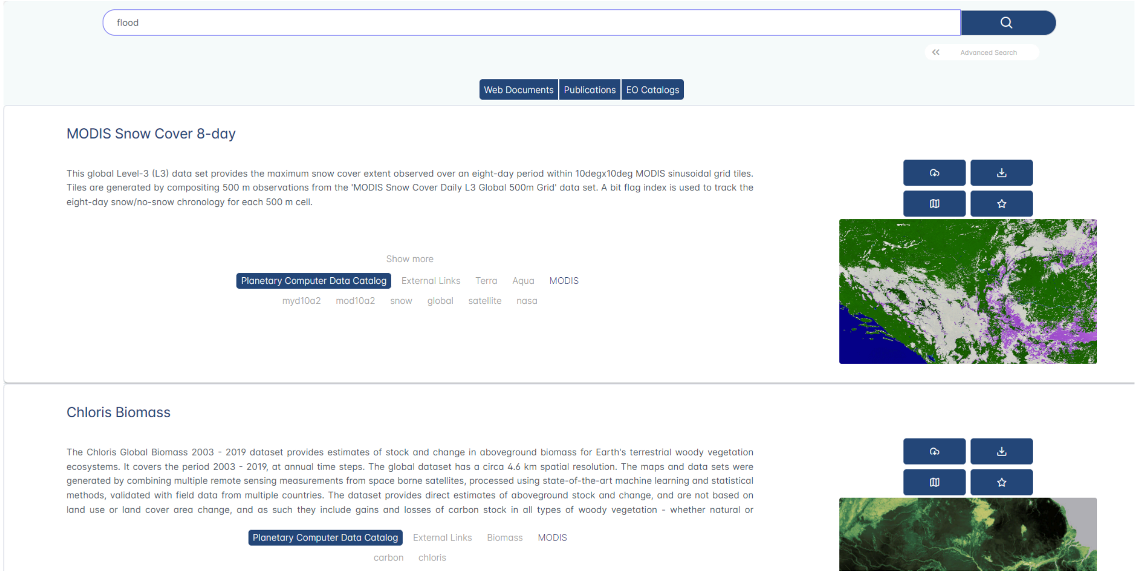
Mosaic Search Applications · Earth Observation Search

Mosaic Vertical Search (TU Graz)
Modular framework for building vertical search engines on OWI slices — domain-specific indices for science, health, arts (8M+ docs)



Demo: mosaic.ows.eu · mosaic.ows.eu/demos

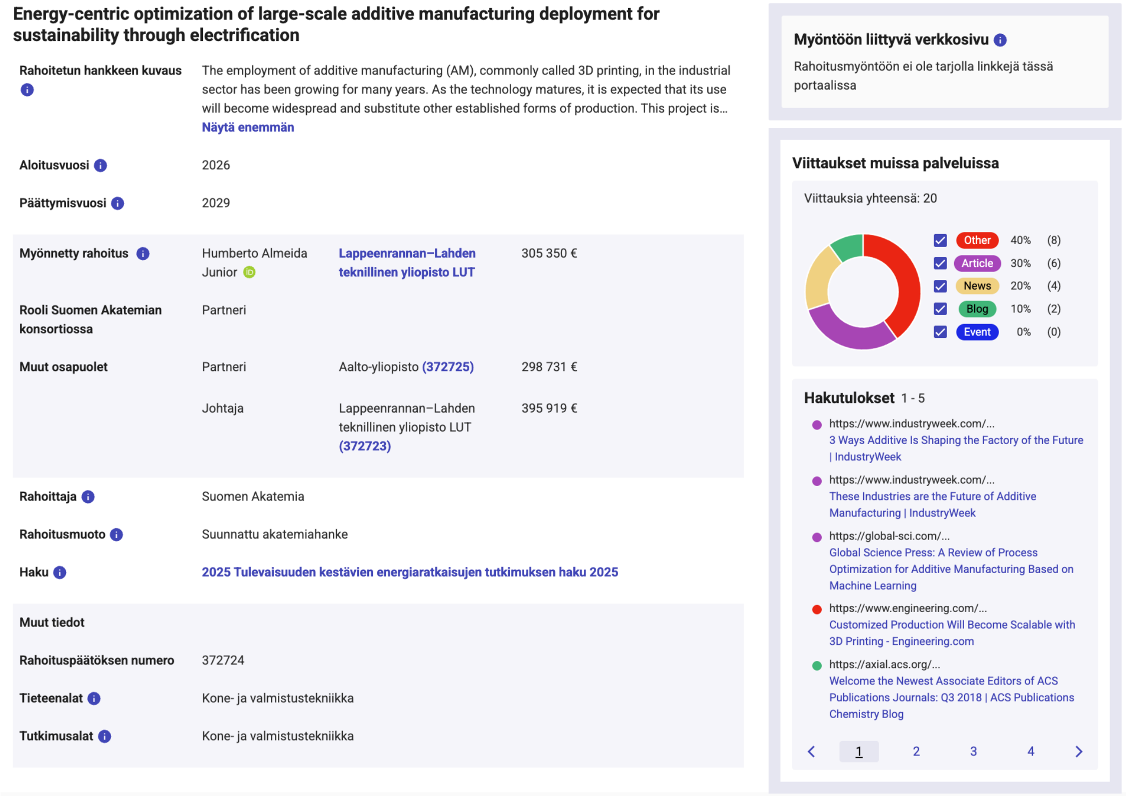
Earth Observation Search (DLR / Radboud)
RAG-based QA over scientific papers, datasets & web content; geo-contextualized multi-genre search on interactive maps; KG with 2M+ nodes (OpenAlex, PANGAEA, DLR Geoservice)



Demo: eo-rag-demo.2.rahtiapp.fi · ows-demo.2.rahtiapp.fi

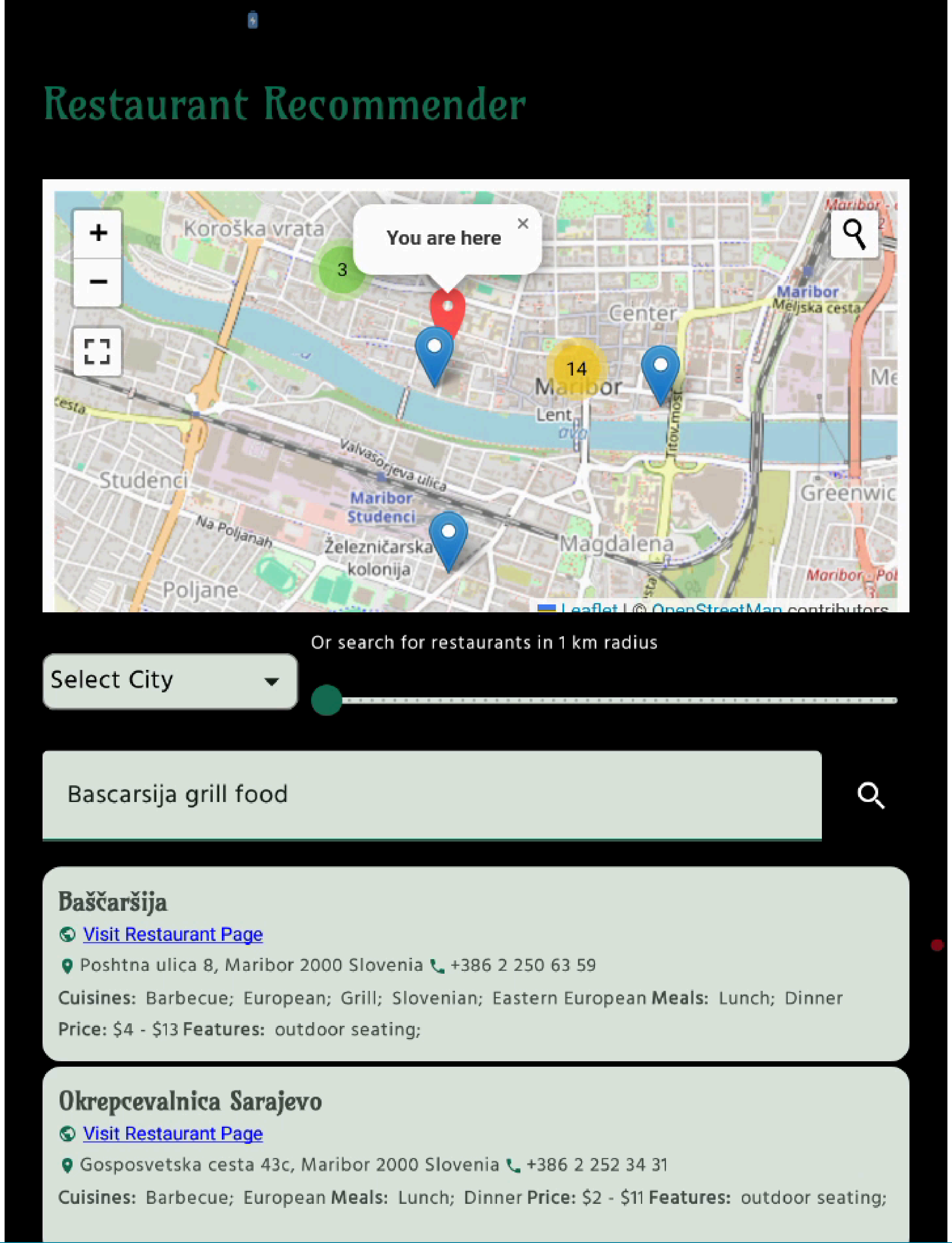
Multilingual Scientific Index · Restaurant Recommender

Multilingual Scientific Index (CSC)
Enriches Finland's Research.fi portal with OWI web data — Finnish & English science content classified by type (publication, blog, event); embedded directly in national research portal



Demo: research.fi

Restaurant Recommender (A1 Slovenia)
Privacy-preserving mobile app; on-device personalization via query imputation; LLM-extracted restaurant features from OWI; 5-month pilot with 214 users — overall satisfaction 4.2 / 5


Restaurant Recommender
You are here
Koroška vrata
Center
Maribor
Lent
Studenci
Maribor Studenci
Železničarska kolonija
Magdalena
Poljane
Valvasorjeva ulica
Na Poljanah
Leaflet | © OpenStreetMap contributors
Or search for restaurants in 1 km radius
Select City
Bascarsija grill food
Baščaršija
Visit Restaurant Page
Poshtna ulica 8, Maribor 2000 Slovenia +386 2 250 63 59
Cuisines: Barbecue; European; Grill; Slovenian; Eastern European Meals: Lunch; Dinner
Price: $4 - $13 Features: outdoor seating;
Okrepcevalnica Sarajevo
Visit Restaurant Page
Gosposvetska cesta 43c, Maribor 2000 Slovenia +386 2 252 34 31
Cuisines: Barbecue; European Meals: Lunch; Dinner Price: $2 - $11 Features: outdoor seating;

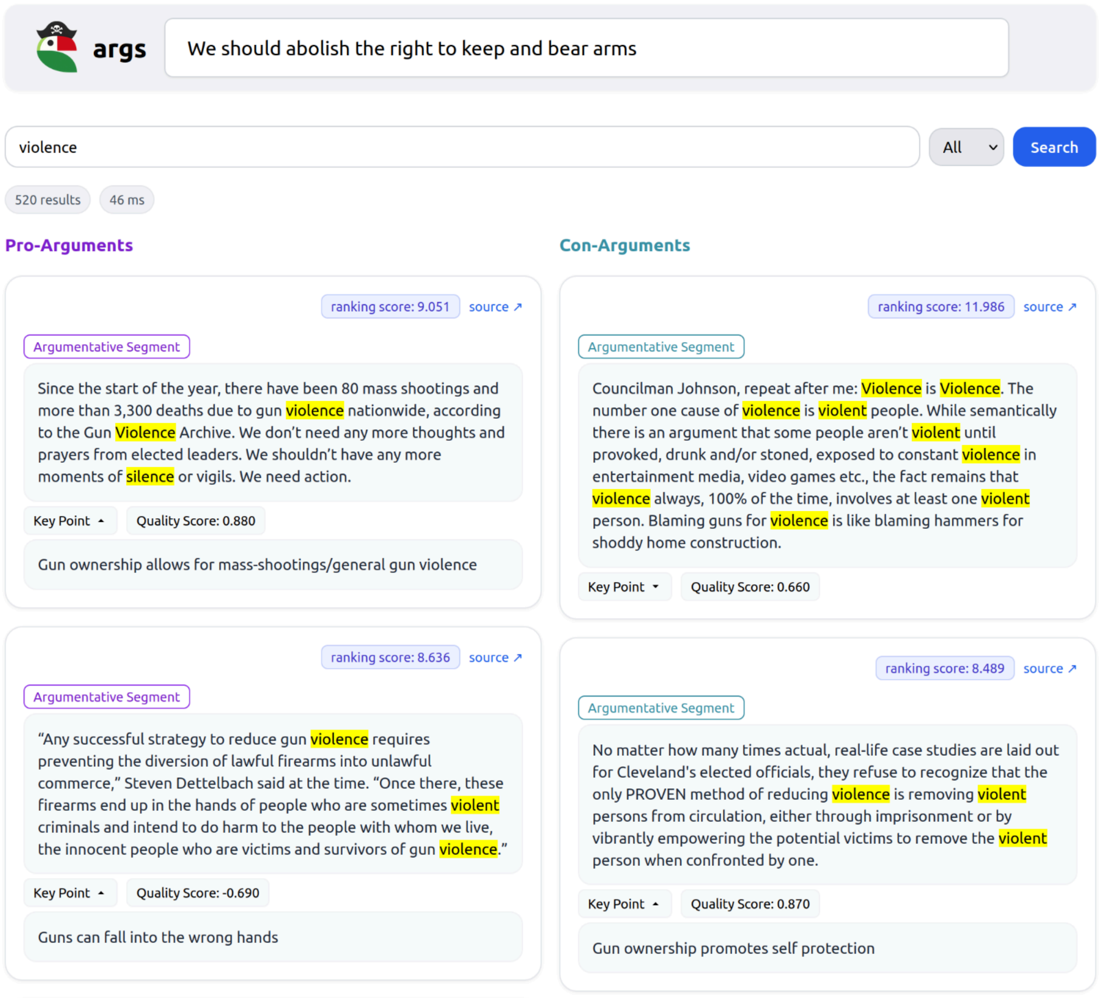
Argumentation Search · CERN Search Applications

Argumentation Search (WEBIS)

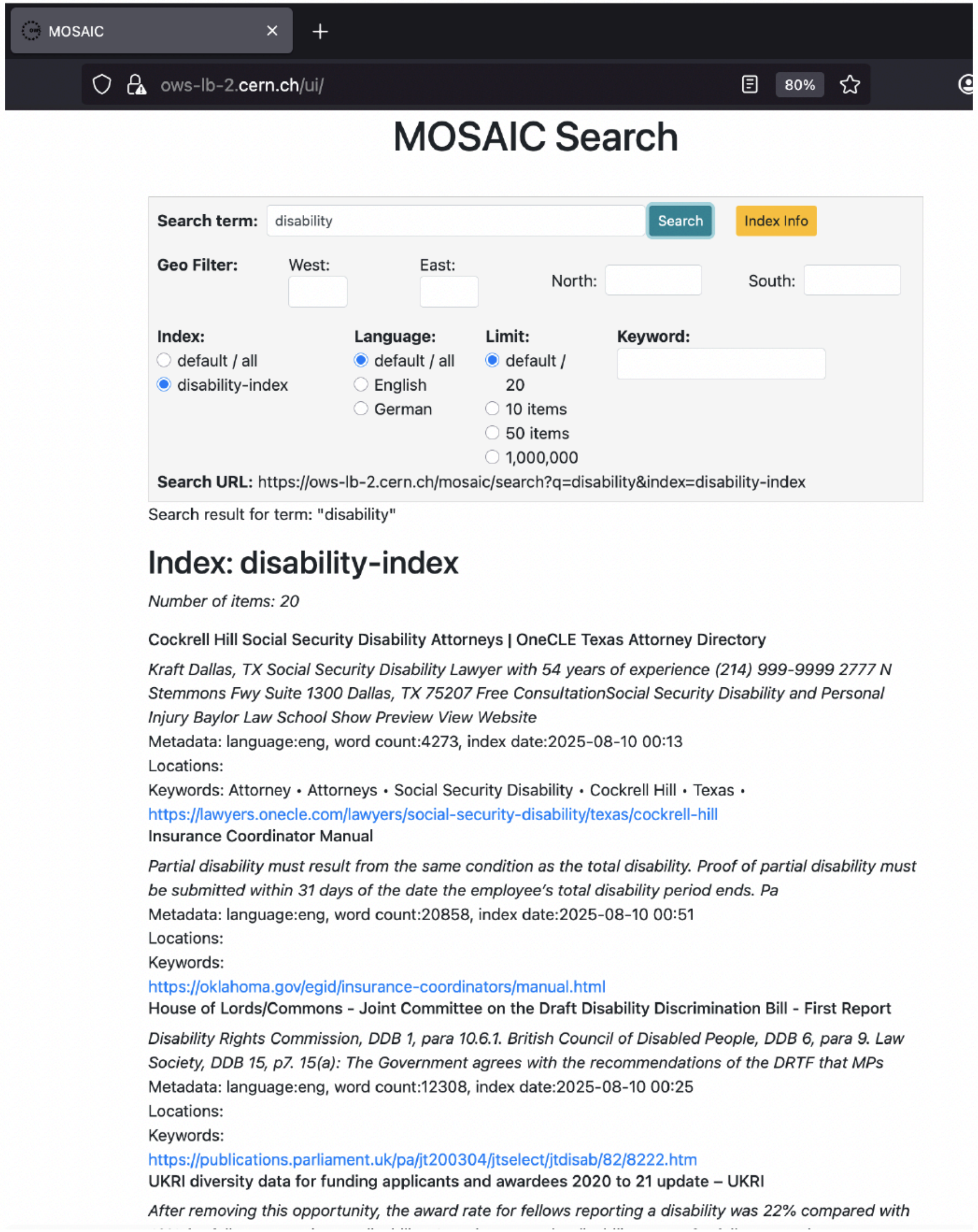
args.me extended with OWI: pro & contra arguments on 31 controversial topics; argument quality scoring, key point matching, temporal trend analysis

CERN Search Applications

- **Institutional**: 25K+ pages crawled from CERN internal web; full OWS pipeline → MOSAIC index for AccGPT knowledge base
- **Nooon**: disability knowledge search (2.59M docs) for HR & Diversity & Inclusion



Demo: args.me/owi · args.me

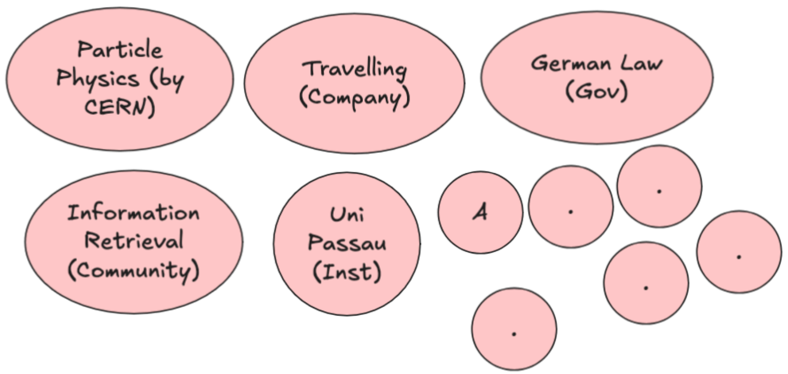


Deployed on open-science-search.cern.ch

Third-Party Projects (FSTPs)

- **AKSE** — argumentation knowledge graph for structured deliberation
- **VERITAS / DEXAI** — verification-oriented RAG for sensitive geopolitical contexts (Ukraine)
- **OMMS** — enriching OpenStreetMap POI with OWI structured data
- **DTCommerce** — product extraction & enrichment pipeline for SME e-commerce
- **CIFFIL** — cross-collection index interoperability for municipal search (Dutch pilots)
- **TILDE** — trustworthy health search with bias detection and entity-level retrieval

Towards Federated RAG Systems



- RAG / Agentic systems require focused, high-quality collections
- Web information redundancy increases due to LLMs — no one-size-fits-all retrieval
- OpenWebSearch.eu supports low-cost creation of topic-specific RAG systems

Key ideas:

- Retrieve from well-designed RAG systems instead of generic search engines
- Topic-specific, deduplicated and curated content
- Domain-optimized ranking, aggregation and summarization
- Lightweight browser-based federation with personal data integration

> **OwlerLite** (WWW 2026) — privacy-respecting browser extension for scope- and freshness-aware retrieval: user-defined **scopes** (trusted sources), semantic

≡

freshness detection via SimHash + vector similarity, Scope Fidelity improved from 0.64 → **0.83** (negligible NDCG loss)

Using the Open Web Index — Data & Analytics

Using the Open Web Index — Data & Analytics

Published Datasets

Dataset	Scale	Use case
WebFAQ	96M Q&A, 75 langs	QA training, cross-lingual IR
WebFAQ 2.0	198M pairs, 108 langs	Dense retrieval, hard negatives
Imprints	5.25M hosts, geocoded	Enterprise analytics, Eurostat
German Commons	154B tokens	German LLM pre-training
OWS-Curlie-2025	20.7M docs + qrels	IR evaluation benchmark

All datasets available at openwebindex.eu/corpora

Analytics Potential

- **Official Statistics**: enterprise data extraction from legal imprints (Eurostat, Destatis)
- **Media Monitoring**: supply chain analysis, news monitoring (JRC)
- **AI Training Data**: domain-specific corpora for LLM fine-tuning
- **Evaluation Benchmarks**: web-scale IR test collections with relevance assessments
- **Trend Analysis**: temporal web monitoring at Petabyte scale

> The OWI is not just for search — it is a data infrastructure for the web-scale analytics ecosystem

IN DETAIL

WebFAQ — Multilingual FAQ from the Web

- **96 million** Q&A pairs across **75+ languages**
- Extracted from structured FAQ markup (JSON-LD, Microdata) in the Open Web Index
- Fields: question, answer, URL, origin, topic, question type
- Parquet format, hosted on Hugging Face

Use cases: multilingual QA training, information retrieval, cross-lingual transfer learning

huggingface.co/datasets/PaDaS-Lab/webfaq

Top languages	Pairs
English	~30M
German	~8M
French	~6M
Spanish	~5M
...	75+ langs

WebFAQ 2.0 — Hard Negatives for Dense Retrieval

- **198 million** FAQ-based Q&A pairs across **108 languages**
- **14.3 million** bilingual aligned QA pairs — largest FAQ-based resource
- **1.25 million** queries with mined hard negatives (20 languages)
- 200 cross-encoder scored negatives per query

Key contributions:

1. Novel web crawling for diverse content extraction
2. Hard negatives for dense retriever training
3. Contrastive Learning & Knowledge Distillation
4. Open datasets and training scripts

Dinzinger, Caspari, Salman, Topi, Mitrović & Granitzer. "WebFAQ 2.0: A Multilingual QA Dataset with Mined Hard Negatives for Dense Retrieval." arXiv:2602.17327, 2025.

Imprints Dataset — Geospatially Grounded Web Curation

- **5.25 million** hosts, **47.8 GB**
- Each website annotated along two dimensions:
  - **Topical classification** via LLM (Foursquare category hierarchy)
  - **Geographic grounding** from legal imprints via geocoding
- **3.08 million** websites geocoded from extracted addresses
- Page types: main page, imprint, contact, about us, privacy, terms

Country	Geocoded	Share
Germany	2,636,913	85.5%
Austria	223,805	7.3%
Switzerland	160,683	5.2%

Top-Level Topic	Share
Business & Professional	41.6%
Community & Government	16.2%
Arts & Entertainment	12.2%

Dinzinger, Rittlinger, Voigt, Gadziomski, Granitzer, Geiß & Taubenböck. Uni Passau & DLR. License: ODC-BY. openwebindex.eu/corpora

German Commons — 154B Tokens for German LLMs

- **154.56 billion tokens**, **35.78 million documents**
- Aggregated from **41 diverse sources** under open licenses
- Quality-filtered, deduplicated, OCR scored

Domain	Tokens
News Commons	72.67B
Cultural Commons	54.49B
Web Commons	19.89B
Political Commons	3.57B
Legal Commons	2.99B
Scientific Commons	0.84B
Economics Commons	0.11B

Gienapp, Schröder, Schweter et al. "The German Commons — 154 Billion Tokens of Openly Licensed Text for German Language Models." arXiv:2510.13996, 2025.

OWS-Curlie-2025 — Web Search Test Collection

- **20.7 million** documents from 6 months of OWI crawl data (March–August 2025)
- English documents from the Curlie subcollection (websites labeled in the Curlie directory)
- Three evaluation splits with student-provided topics and relevance assessments
- Jina V3 embeddings, CIFF indexes for sparse retrieval
- Accessible via `owilix` CLI and `ir-datasets-owi` Python package

Split	Documents	Embeddings	Qrels
`all`	20,746,181	--	--
`radboud-val`	63,639	yes	yes
`radboud`	90,852	yes	--
`jena-kassel`	77,382	yes	--

Using the Open Web Index — Ideas

Potential Application Domains

> Some application scenarios discussed at the start of OpenWebSearch.eu are now in reach:

Web-Analytics for Official Statistics

Eurostat: enterprise information extraction from legal imprints — who is doing business where?

eurostat

Destatis: statistical data extraction from web sources to complement national statistics

DESTATIS Statistisches Bundesamt

Media Monitoring & Intelligence

JRC Media Monitor: monitoring media ecosystems and supply chain signals at European scale *(in start phase)*

JRC EUROPEAN COMMISSION

A Live Use Case: EU Toy Safety & the Open Web Index

> Can the OWI be used for EU product market surveillance? In preparation of the talk I ran a live exploratory analysis using the OUR^{2}S structured-data API plus some AI Agents.

A Live Use Case: EU Toy Safety & the Open Web Index

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"Make an analysis of the OURS API for the OWI corpus: find children's toys and assess how they appear in the dataset — particularly those potentially violating EU regulations."

Prompt used in Jetbrain's Junie AI System with Gemini 3.1 Flash

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The corpus

Schema	Documents
Web pages	310 M
Structured entities	137.5 M
Unique hosts	10 M

Entities Found

- **27 M Products** — largest single type
- 26 M Organisations
- 18 M Articles
- 3.8 M Local Businesses
- 2.8 M Events

→ Products are the single largest entity type — directly usable for market analysis.

A Live Use Case: Products at Scale — Toys in the Index

Pricing & geography

- ~15 M entities carry price data
- **3 M EUR-priced** products → significant EU market coverage
- Also: USD 2.2 M · UAH 1.9 M · BRL 1.2 M

Toy query results (live)

- `"children toy"` → **102,530 products**
- With pricing → **51,384** (50 % have structured prices)
- `"Spielzeug Kinder"` → **7,888** German results

Sample items found

- *"1:24 die-cast metal sports car"* — $90.99
- *"Kids Microscope 1000× Magnifier"* — $59.00
- *"48-teiliges Einhorn Prinzessin Set"* — €8.44

Structured fields available per product

- Name, brand, GTIN/EAN
- Price, currency, availability
- Description, images, ratings

~50 % of toy products include structured price data, reflecting real e-commerce markup adoption.
~92 % of all entities **lack location data** — a significant data-quality finding.

Whole analysis took 20 minutes in parallel to working on this presentation.

A Live Use Case: Conclusion

What the OWI provides today

- Structured Microdata with some features
  - GTIN/EAN presence assessment
  - Brand & manufacturer data
- Completeness and parsing ok, but should be improved

What would be missing for full EU compliance

- CE marking *(not in schema.org)*
- Age suitability / safety warnings
- Material composition
- ...

Could be added to parser

The OWI is a **foundation for such a use case** — Without an aggregated web index, such an analysis would not be possible. Of course you can use commercial search engines for such tasks, but they are not transparent and you don't know what data they use.

Other Research Directions

Next-Generation Retrieval

- Federated RAG across distributed OWI slices and 3rd-party corpora
- Agentic search orchestration for autonomous knowledge discovery
- Multimodal indexing: images, tables, charts alongside text
- Temporal web search: tracking how the web changes over time

AI & LLM Infrastructure

- Domain-specific corpora for sovereign European LLM training
- Retrieval-Augmented Generation benchmarks on OWI data
- Open evaluation infrastructure for IR and NLP research
- AI Agents & Search - actively pursued by Uni Passau
- **Multi-loop LLM orchestration**: classify incoming tasks, represent task types as vectors, and use cross-encoder-style scoring to route or sequence specialised LLM loops for the identified tasks
- **Stacked Kohonen embeddings**: map each sentence token to the top-n matching SOM codebook vectors, use their grid positions as explicit embedding coordinates, and condition the next token's SOM lookup on the previous token's match to form a joint sequence vector
- **Capability allocation across parameter space**: which capabilities — knowledge, reasoning, behaviour — should live in *model parameters* vs. *indexed memory* vs. *interaction traces*? Open design question; envisioned figure maps capability types across the parameter / retrieval / trace space (parked from the TPC26 "Open System Question" slide)

Societal & Policy Applications

- **Information quality monitoring**: detecting coordinated inauthentic behaviour, SEO spam
- **Digital literacy tooling**: transparent, inspectable search for education
- **Regulatory compliance**: evidence-based auditing of search engine behaviour
- **Digital Sovereignty**: European alternative to US-dominated search infrastructure

The OWI is an open infrastructure — we invite researchers, businesses, and public institutions to build on it

We are looking to be part of interesting research projects to advance the OWI.

We hope to build an open ecosystem that puts open, fair and unbiased information exchange and societal values before commercial interest.

Ethical, Legal & Societal Aspects (ELSA)

ELSA Framework — Values, Risks & Mitigation

10 Ethical Values

1. Transparency
2. Safety & Security
3. Openness
4. Responsibility
5. Privacy
6. Justice
7. Trustworthiness
8. Accountability
9. Sustainability
10. Sovereignty / Autonomy

5 Ethical Risks

1. Control of indexed websites & take-down requests — censorship vs. coverage
2. Flawed content in the OWI — false, misleading, offensive material
3. Inaccurate metadata — errors in ingestion or preprocessing
4. Misuse by third parties — privacy violations, unethical applications
5. Unsuitable governance — undemocratic control, security breaches

ELSA Deliverables

Three iterations (D6.1 → D6.2 → D6.3), final version subsumes all. Developed with:

- 3 multi-day ethics workshops (CAIS / Stiftung Mercator)
- 3 legal FSTP projects (LISA, LOREN, ALMASTIC)
- OSF Working Groups on Ethics & Law

6 Stakeholder Groups

End users · Index users · Developers · Researchers & institutions · Content providers · Governance

Legal Challenges & Regulatory Landscape

Regulatory Framework

- **8+ EU regulations** apply: GDPR, Copyright Directive, DSM Directive, DSA, DMA, Data Act, Data Governance Act, AI Act
- **Copyright is the biggest challenge** — impacts crawling, redistribution, and LLM training in Europe
- No guiding court decisions yet — first time an open web index operates in the EU
- DSA classification unclear — OWI does not fit neatly as hosting service, platform, or search engine
- **AI Act**: OWI classified as **"limited risk"** — research exemption applies during project; post-project needs reassessment

Legal FSTP Projects (published on Zenodo)

- **LISA** — Legal Framework for OWI (Matthias Wendland)
- **LOREN** — Legal Open European Web Index (Paul Johannes, Maxi Nebel)
- **ALMASTIC** — Assessing Legal Risks (Kai Erenli, Andreas Gruber)

Key Legal Risks

- Personal data in crawled content (GDPR Art. 6(1)(f) legitimate interest)
- Copyright infringement — images, paywalled articles, plain text
- Liability as digital intermediary
- Illegal content obligations
- LLM training restrictions (no_genai flag)

10 Technical & Organisational Mitigation Measures

#	Measure	Key Mechanism
M1	Website Management	robots.txt opt-out, blacklisting, quality prioritisation
M2	Take-down Management	Dashboard submission, logged removal, blacklist propagation
M3	Content Analysis	Quality scoring, trigger warnings, ad detection, plugin architecture
M4	Access Control	B2ACCESS registration, immediate blocking of misuse
M5	Transparency Guideline	27-aspect guideline for third-party search apps
M6	Data Protection & Privacy	Anonymised queries, minimal user data, reusable model
M7	Safety & Security	Regular security meetings, incident reporting
M8	OWI Licence	Research (default) · Commercial (OWICL) · Data Access (raw WARC)
M9	Legal & Ethical Self-Assessment	Questionnaire at download time, ethics score, signed PDF certificate
M10	Ethical & Legal Advice	Ethics workshops, legal FSTPs, OSF working groups

Three Licence Types

- **Research** — default; included in owilix
- **Commercial (OWICL)** — requires personal contact with coordinator
- **Data Access** — raw WARC; research only, personal request

Open Web Index Licence (OWIL) – Version 1.0

Last Updated: May 8th, 2024

Preamble

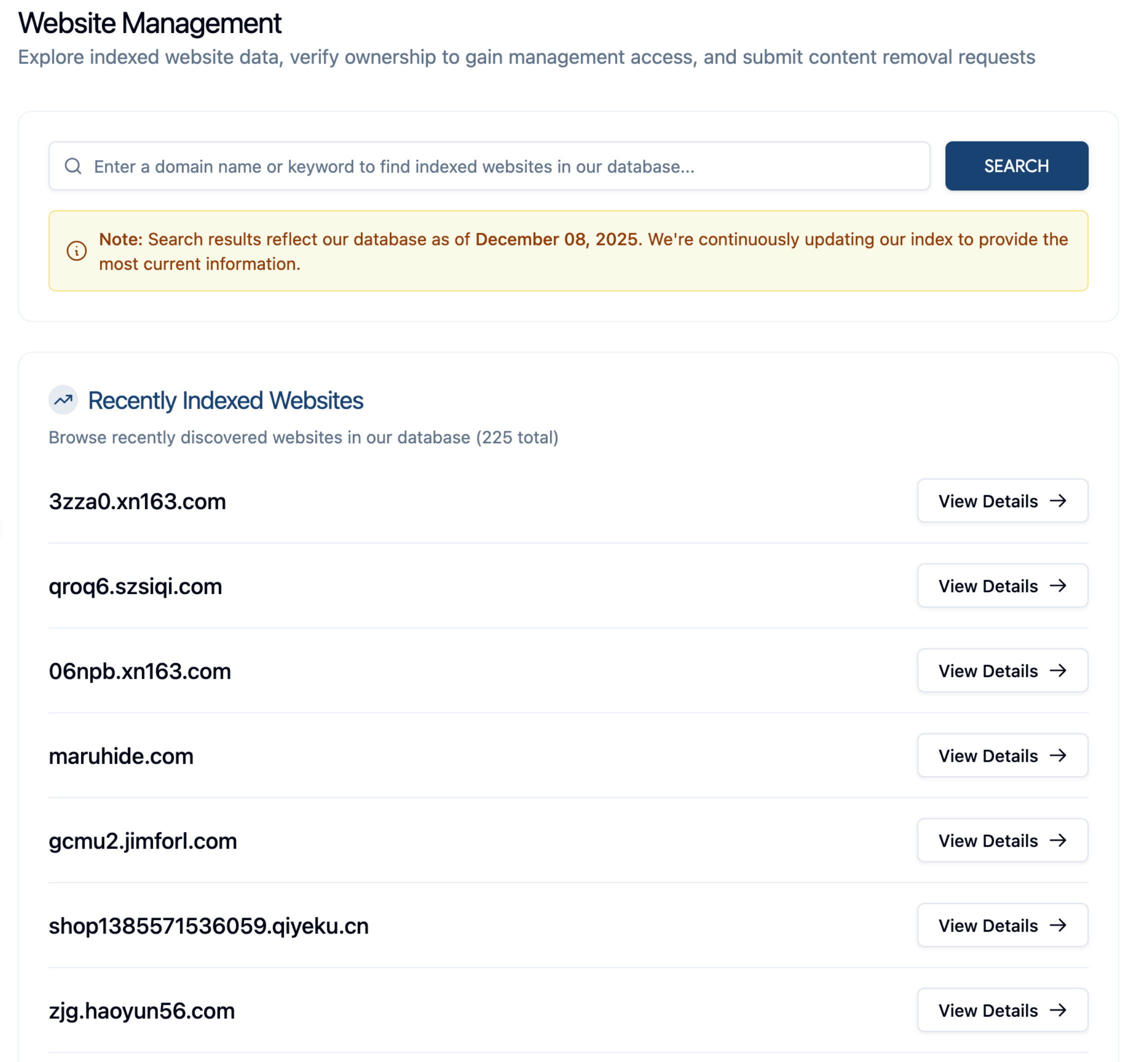
To promote an open human-centred search engine market and provide a solid basis for building front ends for offering a true choice to internet users in selection of preferable search engines, OpenWebSearch.EU develops and pilots the core of a European Open Web Index (OWI). It is the foundation for an open and extensible European Open Web Search and Analysis Infrastructure.

Definitions

"Index Data" – The OpenWebSearch.eu Index Data consists of processed web data derived from previously crawled web data. They are processed in a way that they serve as basis for search applications and web search engines. Hence, they consist of a search index (list of terms and related web documents) and meta data of the original web documents including the plain texts and processed information. The Index Data is distributed over several data centres and organised in partitions.

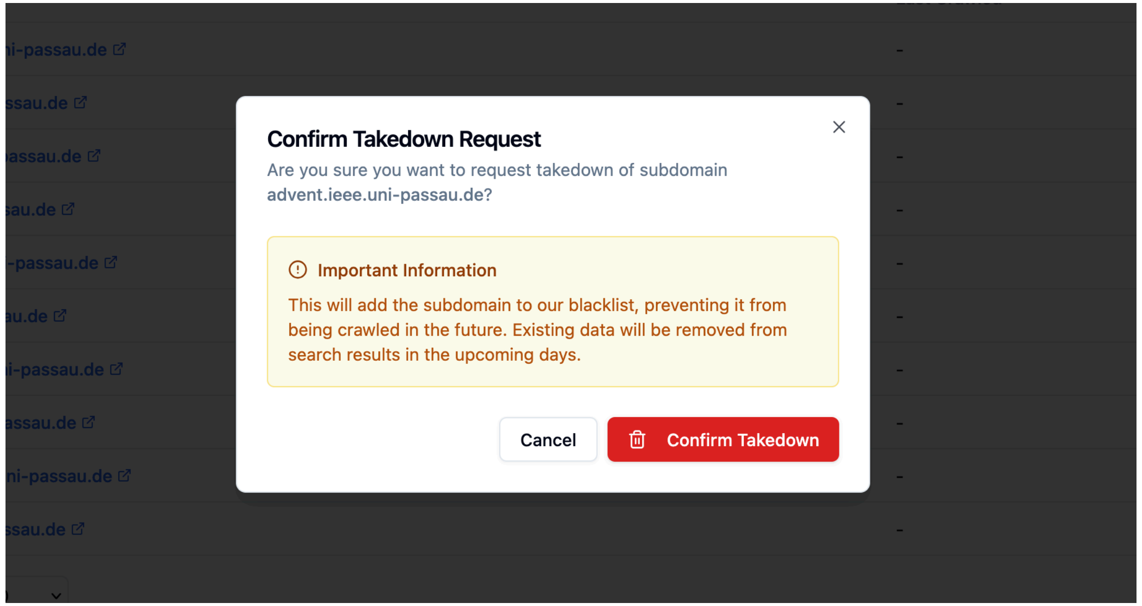
Open Webmaster Console — Transparency Tools

Website Management



- Search any domain in the OWI — crawl stats, content types, language distribution,

Content Removal & Rights

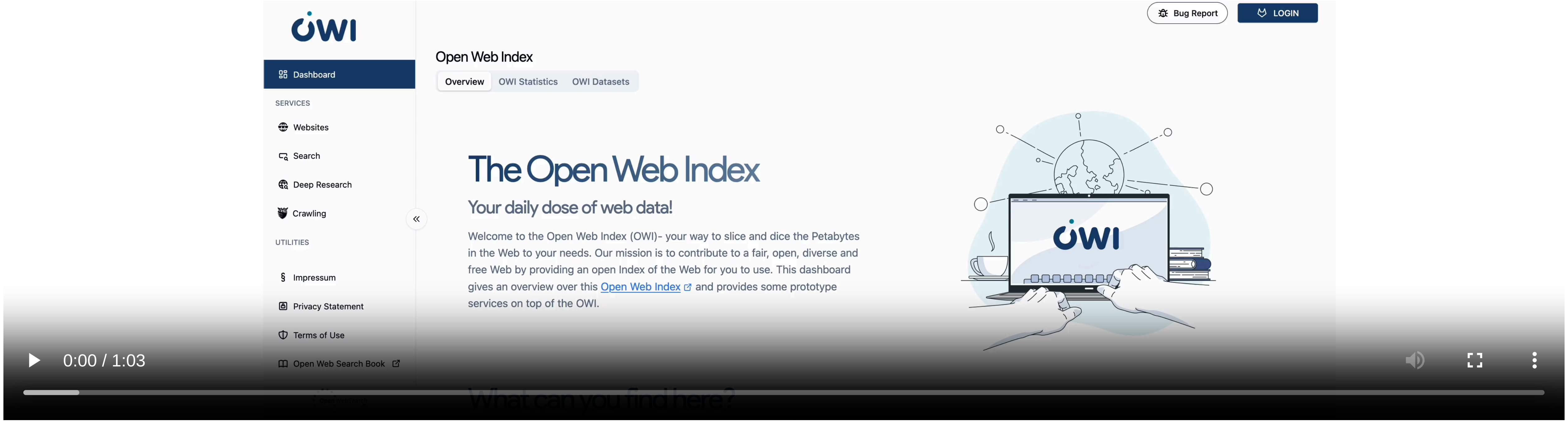


- Verified domain owners can submit **takedown requests** — subdomain or URL level
- Removed content is added to a blacklist; changelog propagated to all index users
- **Ethical self-assessment** survey required before enabling dataset downloads
- **OWLer crawler config**: express crawl constraints directly in the dashboard

subdomains
- **Domain verification**: prove ownership to unlock management capabilities

Transparency & Webmaster Engagement

- Dashboard provides full transparency on crawling activity
- Webmasters can claim ownership of their sites
- Express usage constraints and issue takedown requests
- Gen AI exclusion protocols: robots.txt, user-agent, X-Robots-Tag



FreeWebSearch Charter & Regulatory Reform

#FreeWebSearch Charter — launched 29 September 2025

10 Principles: transparency & explainability · competition & independent infrastructure · privacy without tracking · user/creator/advertiser control · equal access & non-discrimination · information diversity · environmental & social responsibility · integrity against disinformation · search competences & critical awareness · democratic control across borders

Four commitments: (a) serves society, people first; (b) respects democratic values; (c) transparent and open; (d) reflects internet diversity

81 signatories as of February 2026 — openwebsearch.eu/charter

3 Regulatory Reform Proposals

1. **Copyright** — new statutory exception for open web indexes in DSM Directive
2. **Data Protection** — public-interest exemption under Art. 6(1)(e) GDPR
3. **Interoperability** — classify OWI as neutral data intermediary under DSA

Promoted via EU Data Union Strategy consultation (July 2025) and European Parliament breakfast event (December 2025)

Societal Impact

- **Digital sovereignty** — reducing dependency on non-European search engines and AI data pipelines
- **Environmental sustainability** — reducing redundant crawling; shared infrastructure; data centre energy efficiency (CERN heat reuse)
- **Economic opportunity** — ecosystem enabling SMEs and startups; market potential study shows **ROI within 4 years**, projected **net benefit of ~4.5 B EUR over a decade**
- **Democracy & inclusiveness** — unbiased information access for informed citizens; multilingual/multicultural coverage; digital literacy

Proposal: European Web Data Clearing House

A public infrastructure that collects, preprocesses and serves bulk web data as a certified, legally compliant data source — eliminating liability risks for European innovators at scale

Dissemination, Communication & Exploitation

Dissemination, Communication & Exploitation

Dissemination

- **93 papers** published (target: 50) + 11 accepted/submitted
- **3 keynotes**, 10 invited talks, 123 talks & panels
- **#ossym** annual symposium: 2023 · 2024 · 2025 · 2026 (planned)
- WOWS workshop series at ECIR (2024 · 2025 · 2026)
- 3 data challenges · 3 hackathons · 9 webinars
- **11 master theses**, 9 bachelor theses

Community & Stakeholders

Communication

- LinkedIn: **882 followers** (+263%)
- Mastodon: **1,700+ followers** (+353%)
- **63 blogposts** · 12 media articles (5 countries)
- 3 podcast appearances (Deutschlandfunk, Federal Bar)
- **560 newsletter subscribers** (+224%)
- OWI Dashboard: ~1,500 unique visitors/month

Exploitation

Governance

- OSF mandated to coordinate project legacy
- Partners continue under MoU
- Monthly Community Meetup (post-project)
- Target: inclusion in EuroHPC AI Factory infrastructure

Open Source

All code, pipelines, and data formats released as open source; FAIR principles on Zenodo; OWI treated as European public good

- **180+ members** on Mattermost community platform
- #FreeWebSearch Charter: **81 signatories**
- European Parliament breakfast (Dec 2025)
- Web Data Working Group at EuroHPC AI Factories
- Partnership with **EUSP** (Ecosia + Qwant)

- OSF as non-profit sustainability vehicle
- Market potential: **~4.5 B EUR net benefit** over 10 years, ROI within 4 years
- 14 partners with individual follow-on plans
- 20 open-source projects on public GitLab

IN DETAIL

Scientific Dissemination

Publications — Exceeded All Targets

Category	Count
Journal papers	1
Conference & workshop papers	92
Total published	**93**
Accepted (CHIIR 2026)	2
Submitted	9
Grand total	**104**

Target was 50 papers — **186% of goal achieved**

Venues: ECIR, SIGIR, ACL, CLEF, CIKM, EMNLP, WWW, IJCAI, ICTIR etc.

Talks & Events

- 3 keynotes · 10 invited talks · 123 talks, presentations, panels
- 11 workshops · 11 lectures · 5 lectures with tutorials · 1 seminar
- 3 data challenges · 3 hackathons · 9 webinars · 8 training events

Event Series

#ossym — Open Search Symposium

- 2023: inaugural
- 2024: Garching (Munich)
- 2025: CSC Espoo (Helsinki)
- **2026: Berlin** (planned)

WOWS Workshop Series

Web-scale Open Web Search at ECIR — 2024, 2025, 2026 (planned). Also: Dagstuhl Seminar on RAG (Sept 2025)

Theses Supervised

- 11 master theses · 9 bachelor theses

Communication — Organic Growth Across All Channels

Social Media *(organic only — no paid advertising)*

Channel	Followers	Growth
LinkedIn	882	+263% since Feb 2024
Mastodon	1,700+	+353% since Feb 2024

339 posts published since midterm report · avg. 2 posts/week

Media Coverage

- **12 articles** in print/online media across 5 countries (Feb 2024 – Feb 2026)
- Outlets: CERN Courier, c't magazine, heise, tagesspiegel, The Register, HiPEAC magazine, European Perspectives, Bertelsmann Change Quarterly
- **3 podcast appearances**: Besser Wissen Podcast, Deutschlandfunk Kultur, (R)ECHT INTERESSANT (German Federal Bar Association)
- 63 blogposts on project website + ~18 on partner sites

Newsletter & Website

OWI Brand Identity

Complete brand created: logo kit, signet, brand colours, claim

"Time to reclaim the web"

Print Materials

- 1,000 project brochures
- 500 OWI product flyers
- 50 Policy Paper brochures (Parliamentary Breakfast)
- 100 Economy Study copies
- 2m × 3m back-wall display

Community Building & Stakeholder Engagement

Community Platform (Mattermost)

- **180+ members** — public, joinable via openwebsearch.eu/community
- Channels: Tech & Application (most active), Economy, Legal
- Monthly Community Meetup continues post-project (merged Community Updates + OSF WG Tech & Application)

Key Events

- **Public OWI Launch Event** (6 June 2025): 100+ live participants; index state, OWI licence, ethical self-assessment, use cases — recording on Vimeo
- Joint community updates with **NGI Search** (Nov 2024, June 2025)
- Index access training for FSTPs (Nov 2024): owilix + MOSAIC
- Czech-Bavarian Supercomputing Summer School 2025

Policy Engagement

- 26 policy-level meetings · 6 member state briefings
- **European Parliament breakfast** (Dec 2025): MEPs Alexandra Geese (Greens/EFA) + Elena Sancho Murillo (S&D); Deputy DG Renate Nikolay (DG CNECT)
- Policy paper: *"Boosting digital sovereignty and competitiveness through a European Web Data Infrastructure"*
- NGI Forum 2025 · EuroHPC Summit 2025 (Krakow)

Key Partnerships

- **EUSP** (Ecosia + Qwant joint venture) — regular consultation; joint workshop in Paris
- **OpenEuroLLM** — EU sovereign language models; interest in European-language web pages
- **JRC** — company pages for Europe Media Monitor
- **AGES** (Austria) — eMarketShield: online marketplace monitoring for nutritional supplements
- **Eurostat** & national statistical offices — web analytics use cases

EuroHPC Web Data Working Group

Formed Summer 2025, led by LUMI and HammerHAI AI Factories —

advocates for OWI as standard AI Factory infrastructure

Exploitation Strategy & Governance

Market Potential (Mucke Roth & Company assessment)

- **ROI within 4 years** · projected **net benefit ~4.5 B EUR over a decade**
- Business plan modelled: EUR 25k/year per customer, 30 → 356 clients over 4 years; revenue EUR 750k (2026) → EUR 7.6M (2029); break-even at month 24
- Decision: NGO+GmbH model **not pursued** due to regulatory risk and funding uncertainty; governance formalized in D6.9

Post-Project Sustainability

- **OSF** (Open Search Foundation) mandated to coordinate project legacy
- Partners continue operating under **MoU** to bridge until AI Factory inclusion
- New organization with secured long-term funding identified as key need
- 2 new computing centres onboarded: **Exaion** (EDF Group, France) + **Universität Oldenburg** (Germany)

Open-Source Strategy

- 20 open projects on public GitLab
- FAIR principles applied to all Zenodo records
- Top downloaded dataset: Touche23-ValueEval — **37,421 downloads**
- Top downloaded software: OWLer URLFrontier — 210 downloads

Individual Partner Plans

Key follow-on commitments:

- **UNI PASSAU**: BMBF SOURCE (2026), startup incubation, OUR²S
- **BADW-LRZ**: OWS@LRZ, SOURCE (1.5 FTE), EuroHPC WG
- **WEBIS**: Resilipipe, CORAL, Agent CRIS, 3 PhD students
- **TUGraz**: MOSAIC frameworks, 6 master theses
- **CERN**: OWLer continuation, 2–4 PhD positions
- **OSF**: 3 FTE sustained, partnerships with Qwant & Ecosia

Pending regulatory challenges

- WARC demand vs. EU copyright limits on raw data sharing
- Commercial licence still blocked by liability risk
- Legal certainty required for scale-up

Conclusion

In Summary

What we built

- A **federated, open web index** — 1.1 PB raw data, 810+ daily datasets, running 24/7 across European HPC centres
- A **full open-source pipeline** — crawl → preprocess → index → distribute, all reproducible
- A **growing ecosystem** — search apps, RAG systems, datasets, and analytics tools built on the OWI

Why it matters

- Web data and web search are **critical infrastructure for Europe's digital sovereignty**
- The OWI **reduces upfront costs** for tapping the web as resource
- Open, transparent, and FAIR — enabling **reproducible research** and **open innovation**
- **Collaborative** approach to build and maintain the infrastructure

> OpenWebSearch.eu proves that a European open web index is technically feasible, legally navigable, and economically viable

IN DETAIL

Key Messages

- Web-data and web search are **critical for Europe's digital sovereignty**
- **Reducing entry barriers** for researchers, innovators, and businesses
- Opening the core element in web search: the **Open Web Index**
- Open software stack, open pipelines, open data
- Distributed, data-centric, web-scale compute infrastructure
- Web data for **AI and Retrieval Augmented Generation**
- Building an **ecosystem** around data, software, and infrastructure
- Transparency and control for legal, ethical, and societal values

We are looking for ...

- **Researchers & tech innovators** — develop new search & retrieval paradigms and content analysis algorithms
- **Data centres** — help hosting a distributed Open Web Index
- **Industry & business partners** — discover the business models of an Open Web Index
- **Policy makers** — help shaping the governance of an open search ecosystem

Contact

- Email: join@openwebsearch.org
- Community: community@openwebsearch.eu
- Website: openwebsearch.eu
- Dashboard: openwebindex.eu
- Mattermost: openwebsearch.eu/community

Thanks. Questions? Comments? Feedback?

Before we come to the Questions

We are looking for ...

Helping Hands

- **Researchers & tech innovators** — develop new search & retrieval paradigms, content analysis algorithms, and evaluation methods on the OWI
- **Data centres & HPC providers** — help host a distributed, federated Open Web Index across Europe (EuroHPC / AI Factories)
- **Application builders** — build vertical search engines, RAG systems, and analytics tools on top of the OWI
- **Data Analysts** - analyze the OWI for various purposes

Helping Funds

- **Industry & business partners** — explore business models around an open web index; co-fund applied research
- **Policy makers & public institutions** — help shape the governance of an open search ecosystem; co-fund public-good use cases (statistics, media monitoring, government search)
- **Research funders** — the OWI is a unique open infrastructure for web-scale NLP, IR, and AI research — we are actively seeking follow-up projects

Get in touch

Website	openwebsearch.eu
Dashboard	openwebindex.eu
Code Repository	https://opencode.it4i.eu/openwebsearcheu-public/
Community	openwebsearch.eu/community
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General	community@openwebsearch.eu

Thank you — Questions, Comments, Feedback very welcome!